7030 DPS

BX - O
File No. KA EX1
KA EX1A
KA EX1B
KA EX1C

BX - O
BASIC EXCHANGE PARALLEL MAINTENANCE PROGRAM

September 1, 1961

1. Maintenance program. Used to test data paths to and from I/O units and to and from main memory independent of CPU.

2. Programs becoming obsolete. None.

3. KA EX1 and KA EX1A are applicable to systems using 48 ECS printer code. KA EX1B and KA EX1C are applicable to systems using 48 BCD printer code.

TABLE OF CONTENTS

		Page
1.	PURPOSE	1
2.	EQUIPMENT REQUIREMENTS	1
3.	MODES OF CONTROL	2
3.1	Self Control	2
3.1.1	Procedure	2
3.1.2	Success Indications	13
3.1.3	Failure Indications	13
3.1.4	Supplementary Information	14
3.2	DCP Control (Not Applicable)	
4.	PROGRAM PHILOSOPHY	15

1. PURPOSE

The purpose of the BX-0 Maintenance Program is to test data paths to and from the I/O units and to and from Main Memory, independent of CPU.

2. EQUIPMENT REQUIREMENTS

N- Necessary for Basic Testing
A- Additional Requirements for Full Testing
* - Exception

2.1 Testing Requirements

0-8K	8K-16K	16K-32K	32K-Above	Ops Console	Card Reader
		N		N	N

Punch	Printer	Disc	Tapes		
N	N		N		

2.2 Buffer Equipment Requirements

Disc	Tapes

3. MODE OF CONTROL

3.1 Self Control

BX-0 is exclusively a manually operated test, since it is independent of the CPU. It requires the operator to manually set the bits on the exchange maintenance console and manually execute all instructions. The output from each I/O unit test is indicated in the section containing the instructions for that test.

3.1.1 Procedure

I. INITIAL LOAD PROCEDURE

The program can be loaded by normal IPL procedure. If IPL is inoperative, the following can be used:

1. By BX manipulation, place the following CW in a main memory location not used by the program.

   Data Word Address - SLC Value
   Word Count - As shown in the program listing
   Refill - 0
   Chain flag - 0, Multiple flag - 1

2. Read by executing the CW in the location in which it was stored by step 1.

II. OVERALL PROCEDURE

All tests of BX-0 require the operator to manually execute control words in Main Memory and to execute various Control and Locate instructions. The following procedure should be followed in the execution of these instructions:

A. Reading or Writing

1. Place the BX mode switch in the TEST MODE position.

2. Depress CLEAR MEMORY pushbutton.

3. Set "Type of Test" to EX MEM.

4. From the BX-0 listing, obtain the main memory address of control word desired. Place this address in the REFILL ADDRESS of the panel keys.

5. In the EXCHANGE MEMORY ADDRESS switch register, enter the CHANNEL NUMBER desired and bit 128 (Control Word Memory). Make the total parity ODD.

6. Depress the "LOAD MEMORY" switch.

7. Depress SINGLE CYCLE pushbutton twice.

8. Turn OFF the load memory switch.

9. In the exchange memory address switch register, turn OFF bit 128. Parity should now be even.

10. Be sure the channel to be used is NOT blocked by the BLOCK CHANNEL switches. All data word transfer, service request, and channel signal simulation switches should be OFF.

11. Set type of test to Main Memory UNIT.

12. Depress the READ or WRITE pushbutton depending upon instruction desired.

13. Depress the SINGLE CYCLE pushbutton and check ACCEPT response.

14. Depress the START Key. The instruction entered will now be executed.

15. To insure proper operation, stop BX and SINGLE CYCLE through BX control word memory until the channel used is selected. At this time, examine the control word for proper interrupt status bits, data word address, and word count setting. Unless otherwise stated. the normal status bit setting is EOP The flag bits (chain, multiple, and skip) should still be at their original setting.

B. Control or Locate Operations

1. Place the BX Mode switch in the TEST MODE position.

2. Depress CLEAR MEMORY pushbutton.

3. Set "type of test" to UNIT TEST.

4. Set the desired channel number in the EXCHANGE MEMORY ADDRESS switch register, even parity count. (Bit 128 must be OFF.)

5. In the C_0 - C_t panel switches, enter the CONTROL CODE or LOCATE NUMBER desired.

6. Depress the CONTROL or LOCATE pushbutton.

7. Depress SINGLE CYCLE pushbutton and check for ACCEPT response.

8. Depress the START pushbutton and the STOP.

III. INDIVIDUAL TEST PROCEDURE AND OUTPUT

A. Chain Printer Tests

1. Execute the control words as shown on the program listing following the overall test procedure.

2. Check printout for correct data as shown below.

PRT 1 operates with chain, multiple, and skip flags zero. Printout is:

THIS LINE OF PRINT CHECKS THE ABILITY TO PRINT. AB -- YZ12-90%./- #$&*%

PRT 2 operates with multiple flag only set. Printout is an all character print, three lines, each identified. Failure will cause only one line to be printed.

PRT 3 operates with multiple flag set and tests the ability to recognize end codes. Printout is three lines each identified. On failure - All data will be on one line.

PRT 4 tests BX for word count of 1. Printout is WDCT 1- for sucess WDCT 1 FAILURE - on failure

PRT 4A tests BX for word count of 2. The printout is: WORD COUNT 2 - On success WORD COUNT 2 FAILURE - on failure.

PRT 5 operates with multiple and chain flags set. Printout is all data from the above tests, a total of 109 64-bit words.

PRT 6 is a scoping loop which prints the all character print data.

PRT 7 is a scoping loop which prints the end code print data.

PRT 8 is a Suppress Post Spacing test loop. It will suppress post spacing 4 times in each line. For success, all data will be on one line, with normal spacing.

PRT 8 - NOW IS A SUPPRESS POST SPACING TEST LOOP.

PRT 9 is a test of the Select Report functions. It prints according to the Select Report key depressed. If no Select Report keys are depressed all of the select report data will be printed.

The test operates in a continuous loop.

PRT 9 - THIS LINE SHOULD BE PRINTED IF SELECT REPORT 'a' IS DEPRESSED.

Where 'a' corresponds to the Select Report key depressed.

B. Card Reader Tests

1. Place reader test deck in card reader and make reader ready. The test deck is numbered octally in column 80.

2. Execute the control words to read in the test deck.

3. Execute the control words for printout or manually fetch the data and compare.

The first test operates with skip, chain, and multiple flags set. The sequence of data is as follows:

1. One Card Read

   CARD 1 ..... FIRST CARD READ ... DATA IS IN IOS FORMAT. WORD COUNT ON READ WAS 15. READER PATTERNS IN LATER TEST

2. Word Count 1 Test. On success - WDCT1 On failure - WDCT 1 FAILURE IF THIS PRINTS OR IS IN MEMORY WD CNT-1 was not handled by BX .....

3. Word Count 2 Test. On success - WORD COUNT - 2 - On failure - WORD COUNT - 2 - FAILURE

4. Skip Flag Test. On success - THIS IS THE SKIP READ AREA ..... CARD 4 - SKIP FLAG TEST On failure - IF THIS PRINTS SKIP FLAG FAILED.

5. Multiple Flag Test - 3 cards read. Lines of print begin as follows:

   CARD 5
   CARD 6
   CARD 7

   If only one card reads, MF failed and remainder of test will be out of sequence.

6. Long Read Test - 10 cards read. Lines of print begin as follows:

   CARD 8
   CARD 9
   CARD 10
   CARD 11
   CARD 12
   CARD 13
   CARD 14
   CARD 15
   CARD 16
   CARD 17

7. Chain Flag Only Test. For success - CARD 18. TWO CARD READ WITH MF-0. ONLY ONE CARD SHOULD READ .... On failure - THIS CARD SHOULD NOT BE READ CARD 19

C. Tape Unit Tests

1. Execute the control words and control instructions at the proper time by following the program listing.

Since most tape operations require control conditions such as rewind, backspace, etc., the tape test requires that the operator perform these operations from BX following the program listing. Many of the tests are designed specifically to test a particular control function and, therefore, must be run as specified by the program listing. Correct operation is evidenced by the correct printout as shown under the description of each of the five tests.

Test 1. Simple Data and Rewind. Check read-in area manually. Data: An all 1's 8-bit byte shifts left continuously until an all 0's word is reached. Following this an all 1's word, a 101010 word, and a 010101 word.

Test 2. Data and backspace test. Data checked by printing results on the printer. Data follows:

For Success - TEST 2. DATA AND BACKSPACE TEST THIS IS RECORD 1 - TEST TWO .... 10 WORDS, CDSC...

TEST 2. RECORD 2 - 15 WORDS, CDSC .. DATA FOLLOWS --- AB ... YZ01 ... 89 --------
RECORD 3 IS BKSP TEST.

TEST 2. BACKSPACE WORKED IF THIS LINE 3
TEST 2. TEST 2 RECORD 4. 10 WORDS CR.
XXXXXXXXXXX

On Failure - IF THIS PRINTS, BACKSPACE FAILED ..

Test 3. Tape Mark Recognition Test. Data checked by printing results on the printer. Data follows:

For Success - TAPE MARK RECOGNITION RECORD 1.

On Failure - IF THIS PRINTS, TAPE MARK FAILED.

Test 4. Backspace file test. Data checked by printing results on the printer. Data follows:

For Success - TEST 4. BACKSPACE FILE TEST TEST 4.
BACKSPACE FILE TEST PASSED .....

On Failure - TEST 4. BACKSPACE FILE FAILED.

Test 5. Space File Test. Data checked by printing results on printer. Data follows:

For Success - TEST 5. SPACE FILE TEST PASSED.

On Failure - SPACE FILE, TEST 5. FAILED.
XXXXXXXXXXXX

Also included is a sequence of control words which reproduce the program on tape. The tape can then be loaded by IPL procedure.

D. Operator's Console Tests

In the operator's console tests the following is provided.

1. Constants for writing on the console display and typewriter.

2. Reserved locations for reading the console switches and typewriter.

3. Extended typewriter write operations tests.

The procedure for each test follows.

Test 1 and 2. Write Operation

1. Execute the control words - write having the console channel selected.

2. After each control word is executed, check the display for the data indicated.

Test 1. Chain, multiple and skip flags zero.

Word one -
1. Byte number word which numbers the 8-bit bytes left to right 0-7.
2. All 1's word
3. All 0's word
4. Alternate 1's and 0's 8 bit bytes.

Word two - 1. All 8's word

2. All 7's word

3. Blank word

Word three - 1. All 1's word

Test 2. Chain flag set, multiple and skip flags zero.

Chaining two words - Word 1 - Byte pattern
Word 2 - All 8's

Chaining three words - Word 1 - All 1's
Word 2 - All 8's
Word 3 - All 0's

Test 3 and 4. Read Operation

1. Set up data patterns in the console switches and digital pot.

2. Execute the control word to read the switches.

3. Execute the same control word to write the data for checking.

4. Change the patterns and repeat step 2 and 3 for a more complete test.

Test 3. All flag bits zero.

1. Read 1 word
2. Read 2 words
3. Read 3 words

Test 4. Chain flag set.

1. Chains 2 words
2. Chains 3 words

Tests 5, 6 and 7. Typewriter write operation.

1. Execute the control words.

2. Check the printout for correct results.

Test 5. Chain, multiple, and skip flags zero.

1. One word - on success TYP TST
on failure TYP TST FAILED

2. End Code Test - on success - END CODE TEST
on failure - END CODE TEST FAILED

3. One line which is A thru Z 1 thru 0

Test 6. Chain flag set.

1. Chain 2 words - on success - CHAINING TEST S
on failure - FAILED

2. Chain 3 words - on success - CHAINING TEST SUCCESS
on failure - FAILED

Test 7. Chain and Multiple flags set

1. Multiple flag and end code - on success - MLTIPLE TEST SUCCESSFUL
On failure - Spaces between 'MLTPLE' and 'TEST'.

2.. Simultaneous end code and word count zero - on success - MC TST SUCCESS
On failure - Spaces between 'TEST' and 'SUCCESS'.

Test 8 and 9. Typewriter Read Operation

1. Execute control words and read console.

2. Enter data from console typewriter.

3. Using the same control words and write out data for checking.

Test 8. Chain and multiple flags set.

1. Read 40 characters, no flags set.

2. Read 40 characters, chain, read 32 more.

3. Read 8 words multiple flag mode.

4. Read 25 words with multiple flag set.

5. Read 10 words with the multiple flag set, chain, read 8 more words.

Test 9. Chain, multiple and skip flags set.

1. Skip 5 words, read 3 with chain flag only set.

2. Skip 4 words in multiple block mode, chain, read 5 more words.

In the read tests with the multiple flag set, and an end code is entered, the next three words will be read from the console switches.

Typewriter Tests

1. Backspace test loop.

   Loops and types - This is a BACKSPACE test.

2. Ripple test.

   Types 26 lines upper case letters.

3. All character ball movement test.

   Loops and types all characters.

E. Card Punch Tests

Tables of punch formats for checking pattern cards.

1. Non ECC-Mode, 15 words per card- Starting bit position.

Word	Column	Row
1	1	12
2	6	2
3	11	6
4	17	12
5	22	2
6	27	6
7	33	12
8	38	2
9	43	6
10	49	12
11	54	2
12	59	6
13	65	12
14	70	2
15	75	6

2. ECC Mode, 13 words per card.

   All words begin with the C-bits in Row 12

Word	Column
1	1
2	7
3	13
4	19
5	25
6	31
7	37
8	43
9	49
10	55
11	61
12	67
13	73

3. Table of bits on which the ECC bits are based.

ECC Bits	Data Bits
C-0	0-32
C-1	1, 3, 5, ...61, 63, & 32
C2	2-3, 6-7, 10-11, ... 62-63
C-4	4-7, 12-15, ... 60-63
C-8	8-15, 24-31, 40-47, 56-63
C-16	16-31, 48-63
C-32	0, 32-63

C-T is based on overall parity including ECC bits.

Card Punch Test Procedure

1. Make card punch ready.

2. Execute the control words with a write instruction to the card punch.

3. Examine the cards if in the pattern tests, or if in the extended tests use the control words provided for the card reader and printer to check the data.

1. Test 1. Punch Pattern Cards

   Non ECC Mode - Punches a diagonal pattern from Column 1, Row 12, to Column 12, Row 9, a total of 13 cards punched.

   ECC Mode

   1. Punch 9 cards and floats a '1' in the C-bits.

   2. Punch 9 cards and floats a '0' in the C-bits.

   Test 2. Extended Punch Tests

   This test uses printer data and the card reader and chain printer for checking. Each test card is identified with an octal number in the last column.

3.1.2. Success Indications

The success indications are indicated in the detailed test procedure.

3.1.3 Failure Indications

The failure indications are listed in the detailed test procedure.

3.1.4 Supplementary Information

I. Strap Code Control Word Format

The format for a Strap Coded Control Word is as follows:

CW(OP), Data Word Address, Word Count, Refill, where 'OP' is coded as in the table below:

OP	Skip Flag	Multiple Flag	Chain Flag	Operation
CR	0	0	0	Count Within Record
CCR	0	0	1	Chain Counts Within Record
CD	0	1	0	Count Disregarding Record
CDSC	0	1	1	Count Disregarding Record, Skip and Chain
SCR	1	0	0	Skip, Count Within Record
SCCR	1	0	1	Skip, Chain Counts Within Record
SCD	1	1	0	Skip, Count Disregarding Record
SCDSC	1	1	1	Skip, Count Disregarding Record, Skip and Chain

II. Explanation of File Numbers

Four versions of the BX-0 program are presently available. These programs differ only in the printer code used and in the starting location. The versions are:

File No.	Printer Code	Starting Location
KA EX1	48 ECS	50,000
KA EX1A	48 ECS	100,000
KA EX1B	48 BCD	50,000
KA EX1C	48 BCD	100,000

4. PROGRAM PHILOSOPHY

BX-0 is designed for parallel maintenance. It uses control word sequences to test data paths to and from main memory and to and from the I/O units. The test is independent of CPU and requires the ability to get to and from main memory to operate.

All tests start with the simplest control words and proceed to include the chain, multiple and skip flags. The test is executed completely from BX and, therefore does not test communication paths to and from CPU or all of the control functions.

PROGRAM SUMMARY

PROGRAMS OBSOLETED None.

FUNCTION To test the data paths to and from the I/O units and to and from main memory independent of CPU.

BASIC CONTROLS Controlled manually from the BX console.

MANUAL INTERVENTIONS Not applicable.

SUCCESS INDICATIONS Correct data in memory, and correct printouts.

FAILURE INDICATIONS Failure printouts and incorrect data in main memory.

PROGRAM OPTIONS

FIGURE 1

```
SLC,64.0                                                    000100.00
PUNID,KA EX1C                                KA EX1C
END,64.0                                     100.00         000100.00
```

PRNID,BXO - BASIC EXCHANGE OFF LINE MAI.ITENANCE-E.W.JOHNSON

```
PRNS
PUNFUL
@
@
@       E. W. JOHNSON
@
@       SEPTEMBER  1, 1961
        SLC,%8¤77777.0                                                    077777.00
        SEM,6
@
@
        CW%CD¤,START,END-START&1.,0   @IPL CONTROL WORD     100000.00 20 070740.00 00    077777.00
@       THIS CONTROL WORD IS USED TO READ IN PROGRAM
@       AUTOMATICALLY BY NORMAL - INITIAL PROGRAM LOAD -
@       PROCEDURES...IF IPL IS UNAVAILABLE, THE PROGRAM
@       DECK CAN BE MANUALLY READ-IN BY USING THE
@       FOLLOWING PROCEDURE.....
@
@
@          1. BY BX MANIPULATION, PLACE THE FOLLOWING CW
@             IN MAIN MEMORY LOCATION 100.0...
@
@                DATA WD ADR - 7777.0
@                WORD COUNT  -
@                REFILL      - 0
@                CF-0, MF-1
@
@          2. READ BY EXECUTING STORED CW IN LOC. 100.0
@
@       THE FOLLOWING TABLE INDICATES STRAP CONTROL WORD
@       CODING.....
@
@       FORMAT......CW%OP¤,DATA WD ADR,WD COUNT,REFILL
@       ------------------------------------------------------------------
@
@          OP    SKIP   MF   CF   OPERATION
@       ------------------------------------------------------------------
@        CR       0    0    0    COUNT WITHIN RECORD
@        CCR      0    0    1    CHAIN CNTS WITHIN RECORD
@        CD       0    1    0    COUNT DISREGARDING RECORD
@        CDSC     0    1    1    COUNT DISREGARDING RECORD
@                                 SKIP AND CHAIN
@        SCR      1    0    0    SKIP,COUNT WITHIN RECORD
@        SCCR     1    0    1    SKIP,CHAIN COUNTS WITHIN
@                                 RECORD
@        SCD      1    1    0    SKIP,COUNT,DISREGARDING
@                                 RECORD
@        SCDSC    1    1    1    SKIP,COUNT,DISREGARDING
@                                 RECORD,SKIP AND CHAIN
@       ------------------------------------------------------------------
@
@
@       FOR A DETAILED PROGRAM DESCRIPTION, REFER TO
@       PROGRAM WRITE-UP
```

```
START   NOP                     @START OF TEST                          0.30 00                 100000.00
        NOP                     @PRINTER SECTION                        0.30 00                 100000.40
@
@
@       PRINTER TEST CONTROL WORDS
@
PRT1    CW%CR¤,LINE1,17,0       @EXECUTE THIS CONTROL TO TEST           100016.00 00 000420.00 00   100001.00
                                @ABILITY OF PRINTER TO PRINT.
                                @PRINTS ONE LINE OF PRINT INFO.
@
@
PRT2    CW%CD¤,LINE2,51,0       @MF TEST- ALL CHARACTER PRINT.          100037.00 20 001460.00 00   100002.00
                                @NO END CODE- 3 LINES OF PRINT.
@
@
PRT3    CW%CD¤,LINE3,31,0       @MF TEST,END CODE TEST-                 100122.00 20 000760.00 00   100003.00
                                @PRINTS 3 LINES OF PRINT,EACH
                                @IDENTIFIED.
@
@
PRT4    CW%CR¤,BXWC1,1,0        @BX WORD COUNT -1- TEST.-               100161.00 00 000020.00 00   100004.00
                                @USES PRINTER TO INDICATE
                                @SUCCESS,PRINTS WDCT1 ON SUCCESS.
                                @WDCT1  FAILURE-ON FAILURE
@
@
PRT4A   CW%CR¤,BXWC2,2,0        @BX WORD COUNT -2- TEST.                100163.00 00 000040.00 00   100005.00
                                @USES PRINTER TO INDICATE
                                @SUCCESS,PRINTS-WORD COUNT 2-
                                @ON SUCCESS AND-WORD COUNT 2
                                @FAILURE-ON FAILURE.
@
@
PRT5    CW%CDSC¤,LINE1,17,$&1.  @CHAIN FLAG/MULTIPLE FLAG TEST          100016.00 60 000422.00 07   100006.00
        CW%CDSC¤,LINE2,51,$&1.  @DO ALL ABOVE FUNCTIONS                 100037.00 60 001462.00 08   100007.00
        CW%CDSC¤,LINE3,31,$&1.  @WITH CF AND MF SET 1                   100122.00 60 000762.00 09   100010.00
        CW%CDSC¤,BXWC1,1,PRT4A                                          100161.00 60 000022.00 05   100011.00
@
@
PRT6    CW%CDSC¤,LINE2,51,$     @SCOPING LOOP-CONTINOUS PRINT           100037.00 60 001462.00 0A   100012.00
@
PRT7    CW%CDSC¤,LINE3,31,$     @SCOPING LOOP-END CODE PRT              100122.00 60 000762.00 0B   100013.00
@
@
@       SELECT REPORT PRINTER TEST
@
@
        @THE PROGRAM LOOPS PRINTING ACCORDING TO THE
        @SELECT REPORT KEY DEPRESSED.
@
@
@       IF NO KEY IS DEPRESSED THE PROGRAM WILL LOOP
@       PRINTING ALL DATA FROM THIS TEST.
@
@
PRT8    CW%CDSC¤,CCFC,32,$      @LOOP FOR CARRIAGE                      100207.00 60 001002.00 0C   100014.00
                                @CONTROL FIELD TESTS.
@
@
@       SUPPRESS POST-SPACING PRINTER TEST
@
PRT9    CW%CDSC¤,SPS1,17,$      @LOOP FOR SUPPRESS                      100166.00 60 000422.00 0D   100015.00
```

@POST SPACING TEST

END OF PRINTER TESTS

```
*       PRINT DATA
*
*
        CNOP
LINE1   %8¤DD%BU,8,8¤,000        @CHAR CONTROL BYTE                                  000   100016.00
        %  AZ¤DD%BU,8,8¤,THIS LINE OF PRINT CHECKS THE ABILITY TOZ                         100016.10
        %  AZ¤DD%BU,8,8¤, PRINT. ABCDEFGHIJKLMNOPQRSTUVWXZ                                100023.10
        %  AA¤DD%BU,8,8¤,YZ1234567890A                                                    100027.10
        %AZ¤DD%BU,8,8¤,.-%¤#@&$*/,Z                                                      100030.50
        %16¤DD%BU,8,8¤,1A                                                          032   100032.00
        %  AZ¤DD%BU,8,8¤,  ONLY ONE LINE SHOULD PRINT    Z                               100032.10
        %  AZ¤DD%BU,8,8¤,PRT1   Z                                                         100036.10
        CNOP
*
*
LINE2   %8¤DD%BU,8,8¤,000        @CHAR CONTROL BYTE-LINE 1                           000   100037.00
        %  AZ¤DD%BU,8,8¤, ABCDEFGHIJKLMNOPQRSTUVZ                                         100037.10
        %  AT¤DD%BU,8,8¤,WXYZ0123456789 ABCDEFGHIJKLMNOPQT                                100042.00
        %  AQ¤DD%BU,8,8¤,RSTUVWXYZ0123456789 ALL CHARACTEQ                                100046.00
        %  AZ¤DD%BU,8,8¤,R PRINT Z                                                        100052.00
        %AZ¤DD%BU,8,8¤,.-%¤#@&$*/,Z                                                      100053.00
        %16¤DD%BU,8,8¤,1A                                                          032   100054.30
        %  AZ¤DD%BU,8,8¤, THREE LINESZ                                                    100054.40
        %  AZ¤DD%BU,8,8¤, FIRST LINE     Z                                                100056.00
*
        %8¤DD%BU,8,8¤,000        @CHAR CONTROL BYTE-LINE 2                           000   100060.00
        %  AZ¤DD%BU,8,8¤, ABCDEFGHIJKLMNOPQRSTUVZ                                         100060.10
        %  AT¤DD%BU,8,8¤,WXYZ0123456789 ABCDEFGHIJKLMNOPQT                                100063.00
        %  AQ¤DD%BU,8,8¤,RSTUVWXYZ0123456789 ALL CHARACTEQ                                100067.00
        %  AZ¤DD%BU,8,8¤,R PRINT Z                                                        100073.00
        %AZ¤DD%BU,8,8¤,.-%¤#@&$*/,Z                                                      100074.00
        %16¤DD%BU,8,8¤,1A                                                          032   100075.30
        %  AZ¤DD%BU,8,8¤, THREE LINESZ                                                    100075.40
        %  AZ¤DD%BU,8,8¤, SECOND LINE    Z                                                100077.00
*
        %8¤DD%BU,8,8¤,000        @CHAR CONTROL BYTE-LINE 3                           000   100101.00
        %  AZ¤DD%BU,8,8¤, ABCDEFGHIJKLMNOPQRSTUVZ                                         100101.10
        %  AT¤DD%BU,8,8¤,WXYZ0123456789 ABCDEFGHIJKLMNOPQT                                100104.00
        %  AQ¤DD%BU,8,8¤,RSTUVWXYZ0123456789 ALL CHARACTEQ                                100110.00
        %  AZ¤DD%BU,8,8¤,R PRINT Z                                                        100114.00
        %AZ¤DD%BU,8,8¤,.-%¤#@&$*/,Z                                                      100115.00
        %16¤DD%BU,8,8¤,1A                                                          032   100116.30
        %  AZ¤DD%BU,8,8¤, THREE LINESZ                                                    100116.40
        %  AZ¤DD%BU,8,8¤, THIRD LINE     Z                                                100120.00
*
*
LINE3   %8¤DD%BU,8,8¤,000        @CHAR CONTROL BYTE                                  000   100122.00
        %  AZ¤DD%BU,8,8¤,MULTIPLE FLAG EQUAL 1 TEST WITH Z                                100122.10
        %  AZ¤DD%BU,8,8¤,END CODE. THIS IS THE FIRST LINE.....Z                           100126.10
        %8¤DD%BU,8,8¤,376        @FIRST END CODE END OF LINE 1                       376   100132.70
        %8¤DD%BU,8,8¤,000        @CHAR CONTROL BYTE-2ND LINE                         000   100133.00
        %  AZ¤DD%BU,8,8¤,THIS IS THE SECOND LINE OF MF/END CODE TZ                        100133.10
        %  AZ¤DD%BU,8,8¤,EST. 376 IS USED FOR END CODE..Z                                 100140.10
        %8¤DD%BU,8,8¤,376,000                                                      376   100144.00
                                                                                   000   100144.10
        %  AZ¤DD%BU,8,8¤,FAILZ                                                            100144.20
        CNOP
        %8¤DD%BU,8,8¤,000        @CHAR CONTROL BYTE-3RD LINE                         000   100145.00
        %  AZ¤DD%BU,8,8¤,THIS IS THE THIRD AND LAST LINE OF END CZ                        100145.10
        %  AZ¤DD%BU,8,8¤,ODE/MF TEST-PRT3-WD CNT 0 STOPS PRINTZ                           100152.10
        %  AZ¤DD%BU,8,8¤, ON THIS LINE.....Z                                              100156.60
        CNOP
```

```
*
BXWC1   %8¤DD%BU,8,8¤,000                                          000   100161.00
        %  AZ¤DD%BU,8,8¤,WDCT1  Z                                      100161.10
        %  AZ¤DD%BU,8,8¤,FAILUREZ                                     100162.00
*
        CNOP
*
BXWC2   %8¤DD%BU,8,8¤,000                                          000   100163.00
        %  AZ¤DD%BU,8,8¤,WORD COUNT -2- Z                              100163.10
        %  AZ¤DD%BU,8,8¤,FAILURE Z                                     100165.00
*
*       SUPPRESS POST SPACING TEST DATA
*
SPS1    %8¤DD%BU,8,8¤,360,000                                      360   100166.00
                                                                   000   100166.10
        %  AZ¤DD%BU,8,8¤,NOW  Z                                        100166.20
        %8¤DD%BU,8,8¤,376                                          376   100166.70
SPS2    %8¤DD%BU,8,8¤,360,000,000,000,000,000                      360   100167.00
                                                                   000   100167.10
                                                                   000   100167.20
                                                                   000   100167.30
                                                                   000   100167.40
                                                                   000   100167.50
        %  AZ¤DD%BU,8,8¤,IS A SUPPZ                                   100167.60
        %8¤DD%BU,8,8¤,376                                          376   100170.70
SPS3    %8¤DD%BU,8,8¤,360,000,000,000,000,000,000                  360   100171.00
                                                                   000   100171.10
                                                                   000   100171.20
                                                                   000   100171.30
                                                                   000   100171.40
                                                                   000   100171.50
                                                                   000   100171.60
        DD%BU,64,8¤,0                                  0000000000000000000000    100171.70
        %  AZ¤DD%BU,8,8¤,RESS POZ                                     100172.70
        %8¤DD%BU,8,8¤,376,000                                      376   100173.60
                                                                   000   100173.70
SPS4    %8¤DD%BU,8,8¤,360,000,000,000,000,000                      360   100174.00
                                                                   000   100174.10
                                                                   000   100174.20
                                                                   000   100174.30
                                                                   000   100174.40
                                                                   000   100174.50
        DD%BU,64,8¤,0                                  0000000000000000000000    100174.60
        DD%BU,64,8¤,0                                  0000000000000000000000    100175.60
        %  AZ¤DD%BU,8,8¤,ST SPACING T  Z                              100176.60
        %8¤DD%BU,8,8¤,376,000,000,000                              376   100200.40
                                                                   000   100200.50
                                                                   000   100200.60
                                                                   000   100200.70
SPS5    %8¤DD%BU,8,8¤,000                                          000   100201.00
        DD%BU,64,8¤,0                                  0000000000000000000000    100201.10
        DD%BU,64,8¤,0                                  0000000000000000000000    100202.10
        DD%BU,64,8¤,0                                  0000000000000000000000    100203.10
        DD%BU,64,8¤,0                                  0000000000000000000000    100204.10
        %  AZ¤DD%BU,8,8¤,EST LOOP..Z                                  100205.10
        %8¤DD%BU,8,8¤,376                                          376   100206.30
        CNOP                                           0.30 00           100206.40
*
*       SELECT REPORT TEST DATA
*
CCFC    %8¤DD%BU,8,8¤,341,000                                      341   100207.00
                                                                   000   100207.10
        %  AZ¤DD%BU,8,8¤,THIS LINE SHOULD BE PRINTED IF SELECT Z           100207.20
        %  AZ¤DD%BU,8,8¤,REPORT 1 IS DEPRESSED..Z                      100214.00
        %8¤DD%BU,8,8¤,376                                          376   100216.70
        %8¤DD%BU,8,8¤,342,000                                      342   100217.00
```

```
%  AZ¤DD%BU,8,8¤,THIS LINE SHOULD BE PRINTED IF SELECT Z                    000   100217.10
%  AZ¤DD%BU,8,8¤,REPORT 2 IS DEPRESSED..Z                                         100217.20
%8¤DD%BU,8,8¤,376                                                                 100224.00
%8¤DD%BU,8,8¤,344,000                                                       376   100226.70
                                                                            344   100227.00
                                                                            000   100227.10
%  AZ¤DD%BU,8,8¤,THIS LINE SHOULD BE PRINTED IF SELECT Z                          100227.20
%  AZ¤DD%BU,8,8¤,REPORT 3 IS DEPRESSED..Z                                         100234.00
%8¤DD%BU,8,8¤,376                                                            376   100236.70
%8¤DD%BU,8,8¤,350,000                                                       350   100237.00
                                                                            000   100237.10
%  AZ¤DD%BU,8,8¤,THIS LINE SHOULD BE PRINTED IF SELECT Z                          100237.20
%  AZ¤DD%BU,8,8¤,REPORT 4 IS DEPRESSED..Z                                         100244.00
%8¤DD%BU,8,8¤,376                                                            376   100246.70
```

```
                CARD READER TESTS


        *****OPERATOR*****


         PLACE THE READER TEST DECK IN CARD READER
        HOPPER AND MAKE READER READY. THE FIRST CONTROL
        WORD SEQUENCE WILL READ IN THE ENTIRE TEST DECK.


        ....IF IT IS DESIRED TO RUN EACH TEST SEPARATELY,
        THE ENTIRE CONTROL WORD SEQUENCE IS REPEATED WITHOUT
        CHAIN FLAGS. RUN THIS SEQUENCE ONLY IF CHAIN FLAG
        OPERATION IS QUESTIONABLE.ADDITIONAL TESTS ARE
        INCLUDED,SEPARATE TO THE FIRST AND SECOND CW SEQUENCE,
        WHICH CHECK VARIOUS OPTIONS OF READER SUCH AS SCOPING
        FEATURES AND ECC TESTS.....


        ONE TEST DECK IS AVAILABLE FOR THE READER TESTS.

        TEST DECK ONE CONTAINS MOSTLY IQS DATA WHICH ARE
        CHECKED BY EXECUTING CHKRDR CONTROL WORD SEQUENCE
        AND PRINTING RESULTS ON CHAIN PRINTER. THE IQS DATA
        WAS CHOSEN TO BE SELF EXPLANATORY. THE LAST WORD OF
        EACH CARD IS IDENTIFIED AS DESCRIBED BEFORE IN BOTH DECKS

        THE PUNCH TEST OUTPUT CAN ALSO BE USED FOR CHECKING THE
        CARD READER.


RDR     CW%CDSC¤,CARD1,15,$&1.0   @FIRST CARD-IDENTIFIED          100313.00 60 000362.00 A8   100247.00
        CW%CDSC¤,CARD2,1,$&1.0    @SECOND CARD-WORD COUNT 1 TEST. 100332.00 60 000022.00 A9   100250.00
                                  @SHOULD SKIP TO THIRD CARD.
        CW%CDSC¤,CARD3,2,$&1.0    @THIRD CARD-WORD COUNT 2 TEST.  100351.00 60 000042.00 AA   100251.00
                                  @SHOULD SKIP TO FOURTH CARD.
        CW%SCCR¤,CARD4,4,$&1.0    @FIRST 4 WORDS OF CARD4 SHOULD BE 100370.00 50 000102.00 AB  100252.00
                                  @SKIPPED,WITH SKIP FLAG.
        CW%CDSC¤,CARD4&4.0,11,$&1.0  @READ IN REMAINDER OF CARD 4. 100374.00 60 000262.00 AC  100253.00
        CW%CDSC¤,CARD5,45,$&1.0   @READ IN 3 CARDS-MF READ.       100407.00 60 001322.00 AD   100254.00
        CW%CDSC¤,CARD8,150,$&1.0  @LONG READ-10 CARD$.            100464.00 60 004542.00 AE   100255.00
        CW%CR¤,CARD18,30,0        @SHOULD ONLY READ ONE CARD.     100712.00 00 000740.00 00   100256.00
```

```
@
@
@
@       THE RESULTS OF READER TEST CAN EASILY BE DETERMINED BY
@       TWO MEANS.
@
@            1. EXECUTE CHKRDR CNT WDS AND PRINT RESULTS ON
@               CHAIN PRINTER. OR.
@            2. MANUALY FETCH READ IN DATA...THE LAST WORD
@               OF EACH CARD HAS ITS OCTAL CARD NUMBER
@               IN THE LAST 8 BIT POSITIONS.WHERE FULL CARD
@              WAS NOT READ.COMPARE WITH IQS STATEMENTS.
@
@
@       THE ABOVE CONTROL WORD SEQUENCE IS NOW REPEATED
@       WITHOUT CHAIN FLAGS
@
@
RDR1    CW%CR¤,CARD1,15,0          @FIRST CARD                          100313.00 00 000360.00 00     100257.00
@
RDR2    CW%CR¤,CARD2,1,0           @SECOND CARD-WORD COUNT 1 TEST.      100332.00 00 000020.00 00     100260.00
                                   @SHOULD SKIP TO THIRD CARD
@
RDR3    CW%CR¤,CARD3,2,0           @THIRD CARD-WORD COUNT 2 TEST.       100351.00 00 000040.00 00     100261.00
RDR4                               @SHOULD SKIP TO FOURTH CARD.           0.00 00 000000.00 00     100262.00
@
RDR5    CW%SCR¤,CARD4,4,0          @SKIP FIRST 4 WORDS WITH SKIP FLAG.  100370.00 10 000100.00 00     100263.00
@
RDR6    CW%CD¤,CARD4&4,0,11,0      @READ-IN REMAINDER OF CARD 4.        100374.00 20 000260.00 00     100264.00
RDR7    CW%CD¤,CARD5,45,0          @THREE CARD MF READ.                 100407.00 20 001320.00 00     100265.00
@
RDR8    CW%CD¤,CARD8,150,0         @LONG READ- 10 CARDS.                100464.00 20 004540.00 00     100266.00
@
RDR9    CW%CR¤,CARD18,30,0         @SHOULD ONLY READ ONE CARD.          100712.00 00 000740.00 00     100267.00
@
@
@  MORE TESTS WILL BE ADDED AT A LATER DATE
@
@
@
@       THE FOLLOWING GROUP OF CONTROL WORDS PRINT
@       READ IN DATA OF READER TEST. PROVISIONS ARE
@       INCLUDED TO PRINT FAILURE INDICATIONS OF
@       ALL TESTS.FOR EXPLANATION.REFER TO PROGRAM
@       DESCRIPTION WRITE-UP...
@
@
CHKRDR  CW%CDSC¤,CARD1,15,$&1,0                                      100313.00 60 000362.00 B9     100270.00
        CW%CDSC¤,CARD2,15,$&1,0                                      100332.00 60 000362.00 BA     100271.00
        CW%CDSC¤,CARD3,15,$&1,0                                      100351.00 60 000362.00 BB     100272.00
        CW%CDSC¤,CARD4,15,$&1,0                                      100370.00 60 000362.00 BC     100273.00
        CW%CDSC¤,CARD5,15,$&1,0                                      100407.00 60 000362.00 BD     100274.00
        CW%CDSC¤,CARD6,15,$&1,0                                      100426.00 60 000362.00 BE     100275.00
        CW%CDSC¤,CARD7,15,$&1,0                                      100445.00 60 000362.00 BF     100276.00
        CW%CDSC¤,CARD8,15,$&1,0                                      100464.00 60 000362.00 C0     100277.00
        CW%CDSC¤,CARD9,15,$&1,0                                      100503.00 60 000362.00 C1     100300.00
        CW%CDSC¤,CARD10,15,$&1,0                                     100522.00 60 000362.00 C2     100301.00
        CW%CDSC¤,CARD11,15,$&1,0                                     100541.00 60 000362.00 C3     100302.00
        CW%CDSC¤,CARD12,15,$&1,0                                     100560.00 60 000362.00 C4     100303.00
        CW%CDSC¤,CARD13,15,$&1,0                                     100577.00 60 000362.00 C5     100304.00
        CW%CDSC¤,CARD14,15,$&1,0                                     100616.00 60 000362.00 C6     100305.00
        CW%CDSC¤,CARD15,15,$&1,0                                     100635.00 60 000362.00 C7     100306.00
```

```
CW%CDSC□,CARD16,15,$61.0
CW%CDSC□,CARD17,15,$61.0
CW%CDSC□,CARD18,15,$61.0
CW%CD□,CARD19,15,0
```

```
100654.00 60 000362.00 C8    100307.00
100673.00 60 000362.00 C9    100310.00
100712.00 60 000362.00 CA    100311.00
100731.00 20 000360.00 00    100312.00
```

```
*
*       READ IN AREA FOR RDR AND RDR1-RDR9 TESTS
*
*
        CNOP
*
CARD1   DR%BU,64,8¤,14                                          16.00       100313.00
        DR%BU,64,8¤,1           @CARD 1 IDENTITY                 1.00       100331.00
*
CARD2   DR%BU,64,8¤,1           @WORD COUNT 1 DATA               1.00       100332.00
        DR%BU,64,8¤,14          @THIS AREA SHOULD BE BLANK      16.00       100333.00
*
CARD3   DR%BU,64,8¤,2           @WORD COUNT 2 DATA               2.00       100351.00
        DR%BU,64,8¤,13          @THIS AREA SHOULD BE BLANK      15.00       100353.00
*
CARD4   %  AZ¤DD%BU,8,8¤, THIS IS THE SKIP READ AREA.....Z                   100370.00
        DR%BU,64,8¤,10          @CARD 4 DATA                    12.00       100374.00
        DR%BU,64,8¤,1           @CARD 4 IDENTITY                 1.00       100406.00
*
CARD5   DR%BU,64,8¤,14          @CARD 5 DATA                    16.00       100407.00
        DR%BU,64,8¤,1           @CARD 5 IDENTITY                 1.00       100425.00
*
CARD6   DR%BU,64,8¤,14          @CARD 6 DATA                    16.00       100426.00
        DR%BU,64,8¤,1           @CARD 6 IDENTITY                 1.00       100444.00
*
CARD7   DR%BU,64,8¤,14          @CARD 7 DATA                    16.00       100445.00
        DR%BU,64,8¤,1           @CARD 7 IDENTITY                 1.00       100463.00
*
CARD8   DR%BU,64,8¤,14          @CARD 8 DATA                    16.00       100464.00
        DR%BU,64,8¤,1           @CARD 8 IDENTITY %OCTAL¤         1.00       100502.00
*
CARD9   DR%BU,64,8¤,14          @CARD 9 DATA                    16.00       100503.00
        DR%BU,64,8¤,1           @CARD 9 IDENTITY %OCTAL¤         1.00       100521.00
*
CARD10  DR%BU,64,8¤,14                                          16.00       100522.00
        DR%BU,64,8¤,1           @CARD 10 IDENTITY %OCTAL¤        1.00       100540.00
*
CARD11  DR%BU,64,8¤,14                                          16.00       100541.00
        DR%BU,64,8¤,1           @CARD 11 IDENTITY %OCTAL¤        1.00       100557.00
*
CARD12  DR%BU,64,8¤,14                                          16.00       100560.00
        DR%BU,64,8¤,1           @CARD 12 IDENTITY %OCTAL¤        1.00       100576.00
*
CARD13  DR%BU,64,8¤,14                                          16.00       100577.00
        DR%BU,64,8¤,1           @CARD 13 IDENTITY                1.00       100615.00
*
CARD14  DR%BU,64,8¤,14                                          16.00       100616.00
        DR%BU,64,8¤,1           @CARD 14 IDENTITY                1.00       100634.00
*
CARD15  DR%BU,64,8¤,14                                          16.00       100635.00
        DR%BU,64,8¤,1           @CARD 15 IDENTITY                1.00       100653.00
*
CARD16  DR%BU,64,8¤,14                                          16.00       100654.00
        DR%BU,64,8¤,1           @CARD 16 IDENTITY                1.00       100672.00
*
CARD17  DR%BU,64,8¤,14                                          16.00       100673.00
        DR%BU,64,8¤,1           @CARD 17 IDENTITY                1.00       100711.00
*
CARD18  DR%BU,64,8¤,14                                          16.00       100712.00
        DR%BU,64,8¤,1           @CARD 18 IDENTITY                1.00       100730.00
*
CARD19  DR%BU,64,8¤,15          @CARD 19 SHOULD NOT HAVE READ   17.00       100731.00
```

729-IV- TAPE TESTS

BOTH DATA AND TAPE CONTROL ARE CHECKED IN THESE
TESTS. INSTRUCTIONS ARE INCLUDED WITHIN THE TESTS
INDICATING THE TYPE OF CONTROL INSTRUCTION
NEEDED. ITS CODE FOR MANUAL EXECUTION, AND THE
TIME OF WHICH IT SHOULD BE EXECUTED. EACH
STEP OF A PARTICULAR TEST IS NUMBERED BY ORDER
OF EXECUTION.

TEST 1.- SIMPLE DATA AND REWIND.
MANUALLY LOCATE DRIVE.
1.-REWIND TAPE. CONTROL CODE 01011110
2.-EXECUTE FOLLOWING CONTROL WORD-WRITE

CW%CR□,RCRDA,12,0 101004.00 00 000300.00 00 100750.00

3.-REWIND TAPE. CONTROL CODE 01011110
4.-EXECUTE FOLLOWING CONTROL WORD-READ

CW%CR□,TPRD1,12,0 101140.00 00 000300.00 00 100751.00

TO CHECK DATA, CHECK READ IN AREA MANUALLY.
DATA IS IN A SIMPLE FORM. AN ALL ONES BYTE
SHIFTS CONTINUALY TO THE LEFT ONE FULL BYTE
FOR EACH WORD READ UNTILL AN ALL ZEROS WORD
IS REACHED. FOLLOWING THIS IS AN ALL ONES
WORDS, A 10101.....WORD, AND A 01010.....WORD.

TEST 2.- DATA AND BACKSPACE TEST
TEST CHECKED BY PRINTING RESULTS
ON CHAIN PRINTER.

1.-LOCATE DESIRED DRIVE.
2.-REWIND TAPE. CONTROL CODE 01011110
3.-EXECUTE FOLLOWING GROUP OF CONTROL WORDS-WRITE

```
CW%CDSC¤,RCRD1,10,$&1.0          101020.00 60 000242.01 EB    100752.00
CW%CDSC¤,RCRD2,15,$&1.0          101032.00 60 000362.01 EC    100753.00
CW%CR¤,RCRD3,5,0                 101051.00 00 000120.00 00    100754.00
```

4.-BACKSPACE TAPE. CONTROL CODE 01111110
5.-EXECUTE FOLLOWING CONTROL WORD-WRITE

```
CW%CDSC¤,RCRD4,5,$&1.0           101056.00 60 000122.01 EE    100755.00
CW%CR¤,RCRD5,10,0                101063.00 00 000240.00 00    100756.00
```

6.-REWIND TAPE. CONTROL CODE 01011110
7.-EXECUTE FOLLOWING CONTROL WORDS-READ.

```
CW%CDSC¤,TPRD2,10,$&1.0          101154.00 60 000242.01 F0    100757.00
CW%CDSC¤,TPRD3,15,$&1.0          101166.00 60 000362.01 F1    100760.00
CW%CDSC¤,TPRD4,5,$&1.0           101205.00 60 000122.01 F2    100761.00
CW%CR¤,TPRD5,10,$&1.0            101212.00 00 000242.01 F3    100762.00
```

8.-TO CHECK TESTS, USE ABOVE SET OF CONTROL WORDS AGAIN
ONLY THIS TIME, PRINT READ IN AREA 0.1 PRINTER.

```
@       TEST 3. TAPE MARK RECOGNITION TEST.
@
@       1.-LOCATE DESIRED DRIVE.
@       2.-REWIND TAPE. CONTROL CODE 01011110
@       3.-EXECUTE FOLLOWING CONTROL WORD-WRITE
@
        CW%CD¤,RCRD10,5,$61.0     @SHOULD NOT CHAIN.        101075.00 20 000122.01 F4     100763.00
        CW%CR¤,RCRD10,5,0                                   101075.00 00 000120.00 00     100764.00
@
@       4.-WRITE A TAPE MARK. CONTROL CODE 01001111
@       5.-EXECUTE FOLLOWING CONTROL WORD-WRITE.
@
        CW%CR¤,RCRD11,5,0                                   101102.00 00 000120.00 00     100765.00
@
@       6.-REWIND TAPE. CONTROL CODE 01011110
@       7.-EXECUTE FOLLOWING CONTROL WORD-ONLY ONE RECORD
@         -SHOULD READ. TAPE MARK SHOULD CAUSE DISCONNET AT 6TH
@         -WORD.
@
        CW%CD¤,TPRD6,15,0                                   101224.00 20 000360.00 00     100766.00
@
@       8.-EXECUTE ABOVE CW WITH PRINTER WRITE TO OBSERVE RESULTS.
@
@
```

```
@       TEST 4.-BACKSPACE FILE TEST.
@
@       1.-LOCATE DESIRED DRIVE.
@       2.-REWIND TAPE. CONTROL CODE 01011110
@       3.-EXECUTE FOLLOWING CONTROL WORD-WRITE
@
        CW%CD¤,RCRD12,5,0                    101107.00 20 000120.00 00    100767.00
@
@       4.-WRITE A TAPE MARK. CONTROL CODE 01001111
@       5.-EXECUTE FOLLOWING CW - WRITE
@
        CW%CD¤,RCRD13,5,0                    101114.00 20 000120.00 00    100770.00
@
@       6.-BACKSPACE FILE. CONTROL CODE 01111111
@       7.-EXECUTE FOLLOWING CW-WRITE
@
        CW%CD¤,RCRD14,5,0                    101121.00 20 000120.00 00    100771.00
@
@       8.-REWIND TAPE. CONTROL CODE 01011110
@       9.-EXECUTE FOLLOWING CONTROL WORDS-READ.
@
        CW%CD¤,TPRD7,10,0                    101243.00 20 000240.00 00    100772.00
@
@
@       10.-EXECUTE FOLLOWING CW ON-PRINTER--PRINT,
@
        CW%CD¤,TPRD7,10,0                    101243.00 20 000240.00 00    100773.00
@
@
```

```
          TEST 5 SPACE FILE TEST
          1.-LOCATE DESIRED DRIVE.
          2.-REWIND TAPE. CONTROL CODE 01011110
          3.-EXECUTE FOLLOWING CONTROL WORD-WRITE

          CW%CD¤,RCRD15,5,0                                  101126.00 20 000120.00 00     100774.00

          4.-WRITE A TAPE MARK. CONTROL CODE 01001111
          5.-REWIND TAPE. CONTROL CODE 01011110
          6.-SPACE FILE. CONTROL CODE 00111111
          7.-EXECUTE FOLLOWING CW WRITE.

          CW%CD¤,RCRD16,5,0                                  101133.00 20 000120.00 00     100775.00

          8.-REWIND TAPE. CONTROL CODE 01011110
          9.-EXECUTE FOLLOWING CWS-READ.

          CW%CDSC¤,TPRD8,5,$&1.0                             101262.00 60 000122.01 FF     100776.00
          CW%CD¤,TPRD8&5.0,1,0          @SKIP TAPE MARK      101267.00 20 000020.00 00     100777.00

          9A.-EXECUTE FOLLOWING CW-READ

          CW%CD¤,TPRD8&5.0,5,0                               101267.00 20 000120.00 00     101000.00

          10.-EXECUTE FOLLOWING CW ON PRINTER. -WRITE-

          CW%CR¤,TPRD8&5.0,5,0                               101267.00 00 000120.00 00     101001.00
```

```
*       THE FOLLOWING GROUP OF CONTROL WORDS REPRODUCE
*       THIS PROGRAM USING TAPES AS A STORAGE DEVICE.
*
*       1.-LOCATE DESIRED DRIVE
*       2.-REWIND TAPE. CONTROL CODE 01011110
*       3.-EXECUTE FOLLOWING CONTROL WORDS-WRITE
*
        CW%CCR□,IPLCW,1,$&1,0                    101003.00 40 000022.02 03    101002.00
IPLCW   CW%CD□,START,END-START&1.0,0              100000.00 20 070740.00 00    101003.00
*
*       4.-REWIND TAPE. CONTROL CODE 01011110
*
*       TAPE CAN BE USED AS A PROGRAM TAPE.
*       IPL FROM THIS TAPE WILL PRODUCE SAME DATA AS IF
*       BX-0-WERE LOADED FROM CARDS. TO TRUELY TEST TAPE,
*       CLEAR MEMORY AND IPL. RUN PRINTER TEST FOR A
*       DATA TEST.
*
*       ****TO CREATE A NEW BINARY DECK, USE ABOVE
*       CONTROL WORDS ON A PUNCH WRITE.****
        CNOP
```

```
;       TAPE TESTS DATA
;
;
;       TEST 1.
;
RCRDA   %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,377                  000   101004.00
                                                                       000   101004.10
                                                                       000   101004.20
                                                                       000   101004.30
                                                                       000   101004.40
                                                                       000   101004.50
                                                                       000   101004.60
                                                                       377   101004.70
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,377,000                  000   101005.00
                                                                       000   101005.10
                                                                       000   101005.20
                                                                       000   101005.30
                                                                       000   101005.40
                                                                       000   101005.50
                                                                       377   101005.60
                                                                       000   101005.70
        %8¤DD%BU,8,8¤,000,000,000,000,000,377,000,000                  000   101006.00
                                                                       000   101006.10
                                                                       000   101006.20
                                                                       000   101006.30
                                                                       000   101006.40
                                                                       377   101006.50
                                                                       000   101006.60
                                                                       000   101006.70
        %8¤DD%BU,8,8¤,000,000,000,000,377,000,000,000                  000   101007.00
                                                                       000   101007.10
                                                                       000   101007.20
                                                                       000   101007.30
                                                                       377   101007.40
                                                                       000   101007.50
                                                                       000   101007.60
                                                                       000   101007.70
        %8¤DD%BU,8,8¤,000,000,000,377,000,000,000,000                  000   101010.00
                                                                       000   101010.10
                                                                       000   101010.20
                                                                       377   101010.30
                                                                       000   101010.40
                                                                       000   101010.50
                                                                       000   101010.60
                                                                       000   101010.70
        %8¤DD%BU,8,8¤,000,000,377,000,000,000,000,000                  000   101011.00
                                                                       000   101011.10
                                                                       377   101011.20
                                                                       000   101011.30
                                                                       000   101011.40
                                                                       000   101011.50
                                                                       000   101011.60
                                                                       000   101011.70
        %8¤DD%BU,8,8¤,000,377,000,000,000,000,000,000                  000   101012.00
                                                                       377   101012.10
                                                                       000   101012.20
                                                                       000   101012.30
                                                                       000   101012.40
                                                                       000   101012.50
                                                                       000   101012.60
                                                                       000   101012.70
        %8¤DD%BU,8,8¤,377,000,000,000,000,000,000,000                  377   101013.00
```

```
                                                                                                                    000   101013.10
                                                                                                                    000   101013.20
                                                                                                                    000   101013.30
                                                                                                                    000   101013.40
                                                                                                                    000   101013.50
                                                                                                                    000   101013.60
                                                                                                                    000   101013.70
          %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,000                                                             000   101014.00
                                                                                                                    000   101014.10
                                                                                                                    000   101014.20
                                                                                                                    000   101014.30
                                                                                                                    000   101014.40
                                                                                                                    000   101014.50
                                                                                                                    000   101014.60
                                                                                                                    000   101014.70
          %8¤DD%BU,8,8¤,377,377,377,377,377,377,377,377                                                             377   101015.00
                                                                                                                    377   101015.10
                                                                                                                    377   101015.20
                                                                                                                    377   101015.30
                                                                                                                    377   101015.40
                                                                                                                    377   101015.50
                                                                                                                    377   101015.60
                                                                                                                    377   101015.70
          %8¤DD%BU,64,8¤,125252525252525252525252                  125252525252525252525252                         101016.00
          %8¤DD%BU,64,8¤,052525252525252525252525                  052525252525252525252525                         101017.00
@
@
@         TEST 2.
@
RCRD1     %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.                                                  000   101020.00
          %  AZ¤DD%BU,8,8¤,TEST 2. DATA AND BACKSPACE TESTZ                                                               101020.10
          %  AZ¤DD%BU,8,8¤,.THIS IS RECORD 1 - TEST TWO....Z                                                              101024.00
          %  AZ¤DD%BU,8,8¤,10 WORDS, CDSC..Z                                                                              101030.00
@
@
RCRD2     %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.                                                  000   101032.00
          %  AZ¤DD%BU,8,8¤,TEST 2. RECORD 2 - 15 WORDS, CDZ                                                               101032.10
          %  AZ¤DD%BU,8,8¤,SC...DATA FOLLOWS---ABCDEFGHIJKLZ                                                              101036.00
          %  AA¤DD%BU,8,8¤,MNOPQRSTUVWXYZ0123456789--------A                                                              101042.00
          %  AZ¤DD%BU,8,8¤,RECORD 3 IS BCKSP TEST. Z                                                                      101046.00
@
@
RCRD3     %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.                                                  000   101051.00
          %  AZ¤DD%BU,8,8¤,IF THIS PRINTS, BACKSPACE FAILEZ                                                               101051.10
          %  AZ¤DD%BU,8,8¤,D.......Z                                                                                      101055.00
@
@
RCRD4     %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.                                                  000   101056.00
          %  AZ¤DD%BU,8,8¤,TEST 2. BACKSPACE WORKED IF THIZ                                                               101056.10
          %  AZ¤DD%BU,8,8¤,S LINE 3Z                                                                                      101062.00
@
@
RCRD5     %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.                                                  000   101063.00
          %  AZ¤DD%BU,8,8¤,TEST 2. RECORD 4. 10 WORDS, CR.Z                                                               101063.10
          %  AZ¤DD%BU,8,8¤, THIS IS THE LAST RECORD OF TESTZ                                                              101067.00
          %  AZ¤DD%BU,8,8¤, 2..XXXXXXXXXXXZ                                                                               101073.00
@
@         TEST 3.
@
@
RCRD10    %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.                                                  000   101075.00
          %  AZ¤DD%BU,8,8¤,TEST 3. TAPE MARK RECOGNITION.RZ                                                               101075.10
          %  AZ¤DD%BU,8,8¤,ECORD 1.Z                                                                                      101101.00
@
@
RCRD11    %  AZ¤DD%BU,8,8¤,IF THIS PRINTS,TAPE MARK FAILED.Z                                                            101102.00
```

```
%  AZ¤DD%BU,8,8¤,XXXXXXXXZ                                                    101106.00
@
@       TEST4
@
@
RCRD12  %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.          000   101107.00
        %  AZ¤DD%BU,8,8¤,TEST 4.BACKSPACE FILE TEST. RECZ                       101107.10
        %  AZ¤DD%BU,8,8¤,ORD 1...Z                                              101113.00
@
@
RCRD13  %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.          000   101114.00
        %  AZ¤DD%BU,8,8¤,IF THIS PRINTS,BACKSPACE FILE FZ                       101114.10
        %  AZ¤DD%BU,8,8¤,AILED...Z                                              101120.00
@
@
RCRD14  %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.          000   101121.00
        %  AZ¤DD%BU,8,8¤,TEST 4.BACKSPACE FILE TEST PASSZ                       101121.10
        %  AZ¤DD%BU,8,8¤,ED.....Z                                               101125.00
@
@
@       TEST 5.
@
@
RCRD15  %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.          000   101126.00
        %  AZ¤DD%BU,8,8¤,SPACE FILE,TEST 5, FAILED.XXXXXZ                       101126.10
        %  AZ¤DD%BU,8,8¤,XXXXXXXXZ                                              101132.00
@
@
RCRD16  %8¤DD%BU,8,8¤,000       @CHAR CONTROL BYTE FOR PRINTING.          000   101133.00
        %  AZ¤DD%BU,8,8¤,TEST 5. SPACE FILE TEST PASSED.Z                       101133.10
        %  AZ¤DD%BU,8,8¤,........Z                                              101137.00
@
@
```

```
@
@       TAPE TESTS READ IN AREA
@
@
@       TEST 1.
@
@
TPRD1   DR%BU,64,8¤,8       @8 WORDS-ALL ONES BYTES STARTS AT          10.00          101140.00
                            @BYTE 7 AND SHIFTS LEFT ONE BYTE
                            @FOR EACH WORD.
        DR%BU,64,8¤,2       @ALL ZEROS WORD                             2.00          101150.00
                            @ALL ONES WORD
        DR%BU,64,8¤,1       @10101....WORD                              1.00          101152.00
        DR%BU,64,8¤,1       @01010....WORD                              1.00          101153.00
@
@
@       TEST 2.
@
TPRD2   DR%BU,64,8¤,10                                                 12.00          101154.00
TPRD3   DR%BU,64,8¤,15                                                 17.00          101166.00
TPRD4   DR%BU,64,8¤,5                                                   5.00          101205.00
TPRD5   DR%BU,64,8¤,10                                                 12.00          101212.00
@
@
@       TEST 3.
@
@
TPRD6   DR%BU,64,8¤,15                                                 17.00          101224.00
@
@       TEST 4.
@
@
@
TPRD7   DR%BU,64,8¤,15                                                 17.00          101243.00
@
@
@       TEST 5.
@
TPRD8   DR%BU,64,8¤,10                                                 12.00          101262.00
```

```
@
@       CONSOLE TEST
@
@
@       THIS TEST TESTS READ AND WRITE OPERATION
@       OF THE CONSOLE. CONTROL WORDS AND CONSTANS
@       ARE PROVIDED FOR WRITE OPERATIONS- CONTROL
@       WORDS AND RESERVED LOCATIONS FOR READ OPERATIONS
@
@
@            TEST ONE-TESTS WRITE OPERATION ON CNSL LTS
@
@                                 @TESTING WORD ONE
@
CNSL1   CW%CR¤,WORD1,1,0          @WORD ONE-BYTE NUMBER          101356.00 00 000020.00 00    101274.00
        CW%CR¤,WORD1&1.,1,0       @WORD ONE-ALL ONES             101357.00 00 000020.00 00    101275.00
        CW%CR¤,WORD1&2.,1,0       @WORD ONE-ALL ZEROS            101360.00 00 000020.00 00    101276.00
        CW%CR¤,WORD1&3.,1,0       @WORD ONE-ONES AND ZEROS       101361.00 00 000020.00 00    101277.00
@                                 @       BYTE PATTERN
@
@
@                                 @TESTING WORD TWO
@
        CW%CR¤,WORD1&3.,2,0       @WORD TWO-EIGHTS               101361.00 00 000040.00 00    101300.00
        CW%CR¤,WORD1&4.,2,0       @WORD TWO-SEVENS               101362.00 00 000040.00 00    101301.00
        CW%CR¤,WORD1&1.,2,0       @WORD TWO-BLANK                101357.00 00 000040.00 00    101302.00
@
@
@                                 @TESTING WORD THREE
@
        CW%CR¤,WORD1-1.,3,0       @WORD THREE-ALL ONES           101355.00 00 000060.00 00    101303.00
@
@
@            TEST TWO-TESTS CF ON A WRITE OPERATION
@
                                  @CHAINING TWO WORDS
CNSL2   CW%CCR¤,WORD1&3.,1,CNSL2&1.  @WORD ONE-BYTE PATTERN      101361.00 40 000022.02 C5    101304.00
        CW%CR¤,WORD1&4.,1,0       @WORD TWO-ALL EIGHTS           101362.00 00 000020.00 00    101305.00
@
@
@                                 @CHAINING THREE WORDS
@
        CW%CCR¤,WORD1&1.,1,CNSL2&3.  @WORD ONE-ALL ONES          101357.00 40 000022.02 C7    101306.00
        CW%CCR¤,WORD1&4.,1,CNSL2&4.  @WORD TWO-ALL EIGHTS        101362.00 40 000022.02 C8    101307.00
        CW%CR¤,WORD1&2.,1,0       @WORD THREE-ALL ZEROS          101360.00 00 000020.00 00    101310.00
@
@
@            TEST THREE-TESTS READ OPERATION FROM CNSL SWITCHES
@
CNSL3   CW%CR¤,WORD2,1,0          @READ ONE WORD-DATA            101364.00 00 000020.00 00    101311.00
                                  @WILL BE IN WORD 2
        CW%CR¤,WORD2&1.0,2,0      @READ TWO WORDS-DATA           101365.00 00 000040.00 00    101312.00
                                  @WILL BEGIN AT WORD 2&1.0
        CW%CR¤,WORD2&3.0,3,0      @READ THREE WORDS-DATA         101367.00 00 000060.00 00    101313.00
                                  @WILL BEGIN AT WORD 2&3.0
@
@       USE THE SAME CONTROL WORDS AND WRITE
@       OUT DATA FOR CHECKING.
@
@
@            TEST FOUR-TESTS READ OPERATION FROM
                                  @CNSL SW AND CF
```

```
@
@       REPEAT THIS TEST USING SEVERAL ANALOG TO DIGITAL
@       POT SETTINGS....
@
CNSL4   CW%CCR¤,WORD3,1,CNSL4&1  @CHAINING TWO WORD                101372.00 40 000022.02 CD    101314.00
        CW%CCR¤,WORD3&1.,2,0                                       101373.00 00 000040.00 00    101315.00
@
        CW%CCR¤,WORD4,1,CNSL4&3.0  @CHAINING THREE WORDS           101375.00 40 000022.02 CF    101316.00
        CW%CCR¤,WORD4&1.0,1,CNSL4&4.0  @DATA WILL BEGIN AT WORD 4  101376.00 40 000022.02 D0    101317.00
        CW%CCR¤,WORD4&2.0,1,0                                      101377.00 00 000020.00 00    101320.00
@
@       USE THE SAME CONTROL WORDS AND WRITE
@       OUT DATA FOR CHECKING.
@
```

```
@
@          TEST FIVE-TESTS TYPEWRITER WRITE OPERATION
@                          @AND END CODE
@
CNSL5   CW%CR¤,TYPW1-3.0,4,0     @TYPES ONE WORD                  101400.00 00 000100.00 00   101321.00
                                 @WHICH IS,CR,TYP TST
        CW%CR¤,TYPW2-3.0,5,0     @END CODE TEST-TYPE              101405.00 00 000120.00 00   101322.00
                                 @TWO WORDS AND END
                                 @WORDS ARE, CR,END
                                 @CODE TEST,END
@
        CW%CR¤,TYPW3-3.0,14,0    @TYPE ONE LINE                   101413.00 00 000340.00 00   101323.00
                                 @WHICH IS-
                                 @CR, A B C D E F G
                                 @H I J K L M N O
                                 @P Q R S T U V W X
                                 @Y Z...  1 2 3 4 5 6
                                 @7 8 9 0  BS END CR
@
@
@          TEST SIX-TESTS TYPEWRITER WRITE
                           @OPERATION AND CF
@
CNSL6   CW%CCR¤,TYPW4-3.0,4,CNSL6&1.  @CHAINS TWO WORDS           101431.00 40 000102.02 D5   101324.00
        CW%CR¤,TYPW4&2,1,0       @WORDS ARE-CHAINING              101434.02 00 000020.00 00   101325.00
                                 @TEST S,ON FAILURE-FAIL
@
        CW%CCR¤,TYPW4-3.,4,CNSL6&3.  @CHAINS THREE WORDS           101431.00 40 000102.02 D7   101326.00
        CW%CCR¤,TYPW4&2.,1,CNSL6&4.  @WORDS ARE-CHAINING           101436.00 40 000022.02 D8   101327.00
        CW%CR¤,TYPW4&4.,1,0      @TEST SUCCESS...ON               101440.00 00 000020.00 00   101330.00
@                                @FAILURE-FAIL
@
@          TEST SEVEN-TESTS TYPWRITER WRITE
                           @OPERATION MF AND CF
@
CNSL7   CW%CD¤,TYPW5-3.,9,0      @WRITE THREE WORDS               101441.00 20 000220.00 00   101331.00
@                                @WITH END CODE
@                                @BETWEEN WORDS
@                                @WORDS ARE-MLTPLE
@                                @TEST SUCCESSFUL
@                                @ON FAILURE-FAIL
@
        CW%CDSC¤,TYPW6-3.,4,CNSL7&2.  @WRITE TWO WORDS ON TYPEWRITER 101452.00 60 000102.02 DB   101332.00
        CW%CR¤,TYPW6&1.,4,0      @THE END CODE AND COUNT ZERK     101456.00 00 000100.00 00   101333.00
                                 @OCCUR SIMULTANEOUSLY
@
@
@          TEST EIGHT-TESTS TYPEWRITER
                           @READ OPERATION
@
@       THE FOLLOWING CWS READ 40 CHARACTERS TYPED IN
@
CNSL8   CW%CR¤,TYPR1,8,0                                          101465.00 00 000200.00 00   101334.00
@
@       THE FOLLOWING CWS READ 40 CHARACTERS TYPED IN---
@       CHAINS AND READS 32 MORE..
@
        CW%CCR¤,TYPR2,8,$&1.                                      101475.00 40 000202.02 DE   101335.00
        CW%CR¤,TYPR3,4,0                                          101505.00 00 000100.00 00   101336.00
@
@       USE THE SAME CONTROL WORDS AND WRITE
@       OUT DATA FOR CHECKING.
```

```
*
*
*       THE FOLLOWING CWS TEST MF AND CF
*       WHEN IN MF MODE AND AN END CODE IS
*       ENTERED FROM THE CONSOLE TYPEWRITER THE
*       NEXT 3 WORDS WILL BE READ FROM CNSL SWITCHES.
*
*       FOR ONE TEST-COUNT CHARACTERS AND
*       HAVE THE END CODE AND COUNT ZERO OCCUR
*       SIMULTANEOUSLY.....
*
        CW%CD¤,TYPR4,8,0          @READ IN MF MODE          101511.00 20 000200.00 00    101337.00
        CW%CD¤,TYPR5,25,0         @READ IN MF MODE          101521.00 20 000620.00 00    101340.00
        CW%CDSC¤,TYPR7,10,&-2.    @MF AND CF SIM-TYPE 8 CHAR 101556.00 60 000257.77 FE   101341.00 C
*
        CW%CDSC¤,TYPR8,20,CNSL8&8.0  @MORE MF AND CF CW     101570.00 60 000502.02 E4    101342.00
        CW%CCR¤,TYPR9,20,CNSL8&9.0                          101614.00 40 000502.02 E5    101343.00
        CW%CR¤,TYPR10,20,0                                  101640.00 00 000500.00 00    101344.00
*
*
*             TEST NINE -TESTS READ OPERATION
*                              @WITH SF,MF,AND CF.
*
        @THE FOLLOWING CWS ARE FOR READING
        @WITH MF, SF, AND CF.
*
*       EXECUTE THE FOLLOWING CW TO TEST CF AND SF.
*
CNSL9   CW%SCCR¤,TYPR11,5,CNSL9&1.0  @SF AND CF TEST, SKIP 5  101664.00 50 000122.02 E6    101345.00
*
        CW%CR¤,TYPR12,3,0         @TYPE 3 WORDS             101667.00 00 000060.00 00    101346.00
*
*       CW TO PRINT OUT DATA ON CONSOLE.
*
*
        CW%CR¤,TYPR12-3.,6,0                                101664.00 00 000140.00 00    101347.00
        CW%CR¤,TYPR11,5,0         @CW FOR TEST SF AND CF    101664.00 00 000120.00 00    101350.00
*
*
*       EXECUTE THE FOLLOWING CW TO TEST CF, SF, AND MF.
*
        CW%SCDSC¤,TYPR13,4,$&1.       @SF CF, AND MF TEST   101672.00 70 000102.02 EA    101351.00
        CW%CD¤,TYPR14,5,0         @DISREGARDS END CODES     101676.00 20 000120.00 00    101352.00
*
*       CW TO PRINT OUT DATA ON CONSOLE.
*
        CW%CD¤,TYPR13,4,0         @CW FOR TEST SF,CF, AND MF 101672.00 20 000100.00 00   101353.00
        CW%CD¤,TYPR14-3.,6,0                                101673.00 20 000140.00 00    101354.00
*
```

```
                .
                DR%BU,64,8¤,1                                                   1.00                    101355.00
WORD1           %8¤DD%BU,8,8¤,000,001,002,003,004,005,006,007  @BYTE NUMBER WD                  000     101356.00
                                                                                                001     101356.10
                                                                                                002     101356.20
                                                                                                003     101356.30
                                                                                                004     101356.40
                                                                                                005     101356.50
                                                                                                006     101356.60
                                                                                                007     101356.70
                %8¤DD%BU,8,8¤,377,377,377,377,377,377,377,377  @ALL ONES WORD                   377     101357.00
                                                                                                377     101357.10
                                                                                                377     101357.20
                                                                                                377     101357.30
                                                                                                377     101357.40
                                                                                                377     101357.50
                                                                                                377     101357.60
                                                                                                377     101357.70
                DD%BU,64,8¤,0                                   @ALL ZEROS WORD     000000000000000000000   101360.00
                %8¤DD%BU,8,8¤,377,000,377,000,377,000,377,000   @BYTE PATTERN                   377     101361.00
                                                                                                000     101361.10
                                                                                                377     101361.20
                                                                                                000     101361.30
                                                                                                377     101361.40
                                                                                                000     101361.50
                                                                                                377     101361.60
                                                                                                000     101361.70
                %8¤DD%BU,8,8¤,210,210,210,210,210,210,210,210  @ALL EIGHTS                      210     101362.00
                                                                                                210     101362.10
                                                                                                210     101362.20
                                                                                                210     101362.30
                                                                                                210     101362.40
                                                                                                210     101362.50
                                                                                                210     101362.60
                                                                                                210     101362.70
                %8¤DD%BU,8,8¤,167,167,167,167,167,167,167,167  @ALL SEVENS                      167     101363.00
                                                                                                167     101363.10
                                                                                                167     101363.20
                                                                                                167     101363.30
                                                                                                167     101363.40
                                                                                                167     101363.50
                                                                                                167     101363.60
                                                                                                167     101363.70
                .
WORD2           DR%BU,64,8¤,6                                   @READ OPERATION     6.00                101364.00
WORD3           DR%BU,64,8¤,3                                   @DATA RESERVATIO    3.00                101372.00
WORD4           DR%BU,64,8¤,3                                                       3.00                101375.00
TYPW0           DR%BU,64,8¤,3           @RESERVES LOCATIONS FOR                     3.00                101400.00
                                        @FIRST THREE WORDS IN
                                        @A TYPEWRITER OPERATION
TYPW1           %16¤DD%BU,8,8¤,FD,53,5D,4B,00,53,51,53  @CR,TYP TEST                        375     101403.00
                                                                                                123     101403.10
                                                                                                135     101403.20
                                                                                                113     101403.30
                                                                                                000     101403.40
                                                                                                123     101403.50
                                                                                                121     101403.60
                                                                                                123     101403.70
                %16¤DD%BU,8,8¤,37,2D,3D,43,35,33,00,00  @FAILED                                 067     101404.00
                                                                                                055     101404.10
                                                                                                075     101404.20
                                                                                                103     101404.30
```

```
                DR%BU,64,8¤,3              @DATA RESERVATION                    3.00
TYPW2           %16¤DD%BU,8,8¤,FD,35,47,33,00,31,49,33  @END COD
                %16¤DD%BU,8,8¤,35,00,53,35,51,53,00,FE  @F TEST, END
                %16¤DD%BU,8,8¤,37,2D,3D,43,35,33,00,00  @FAILED
                DR%BU,64,8¤,3              @DATA RESERVATION                    3.00
TYPW3           %16¤DD%BU,8,8¤,FD,2D,00,2F,00,31,00,33  @CR, A B C D
                %16¤DD%BU,8,8¤,00,35,00,37,00,39,00,3B      @E F G H
                %16¤DD%BU,8,8¤,00,3D,00,3F,00,41,00,43      @I J K L
                %16¤DD%BU,8,8¤,00,45,00,47,00,49,00,4B      @M N O P
                %16¤DD%BU,8,8¤,00,4D,00,4F,00,51,00,53      @Q R S T
```

```
055  101404.40
063  101404.50
000  101404.60
000  101404.70
     101405.00
375  101410.00
065  101410.10
107  101410.20
063  101410.30
000  101410.40
061  101410.50
111  101410.60
063  101410.70
065  101411.00
000  101411.10
123  101411.20
065  101411.30
121  101411.40
123  101411.50
000  101411.60
376  101411.70
067  101412.00
055  101412.10
075  101412.20
103  101412.30
065  101412.40
063  101412.50
000  101412.60
000  101412.70
     101413.00
375  101416.00
055  101416.10
000  101416.20
057  101416.30
000  101416.40
061  101416.50
000  101416.60
063  101416.70
000  101417.00
065  101417.10
000  101417.20
067  101417.30
000  101417.40
071  101417.50
000  101417.60
073  101417.70
000  101420.00
075  101420.10
000  101420.20
077  101420.30
000  101420.40
101  101420.50
000  101420.60
103  101420.70
000  101421.00
105  101421.10
000  101421.20
107  101421.30
000  101421.40
111  101421.50
000  101421.60
113  101421.70
000  101422.00
115  101422.10
000  101422.20
```

```
                                                                                117   101422.30
                                                                                000   101422.40
                                                                                121   101422.50
                                                                                000   101422.60
                                                                                123   101422.70
        %16¤DD%BU,8,8¤,00,55,00,57,00,59,00,5B      @U V W X                    000   101423.00
                                                                                125   101423.10
                                                                                000   101423.20
                                                                                127   101423.30
                                                                                000   101423.40
                                                                                131   101423.50
                                                                                000   101423.60
                                                                                133   101423.70
        %16¤DD%BU,8,8¤,00,5D,00,5F,74,74,74,00      @Y Z...                     000   101424.00
                                                                                135   101424.10
                                                                                000   101424.20
                                                                                137   101424.30
                                                                                164   101424.40
                                                                                164   101424.50
                                                                                164   101424.60
                                                                                000   101424.70
        %16¤DD%BU,8,8¤,00,62,00,64,00,66,00,68      @1 2 3 4                    000   101425.00
                                                                                142   101425.10
                                                                                000   101425.20
                                                                                144   101425.30
                                                                                000   101425.40
                                                                                146   101425.50
                                                                                000   101425.60
                                                                                150   101425.70
        %16¤DD%BU,8,8¤,00,6A,00,6C,00,6E,00,70      @5 6 7 8                    000   101426.00
                                                                                152   101426.10
                                                                                000   101426.20
                                                                                154   101426.30
                                                                                000   101426.40
                                                                                156   101426.50
                                                                                000   101426.60
                                                                                160   101426.70
        %16¤DD%BU,8,8¤,00,72,00,60,00,00,FC,FD      @9 0  BS, CR                000   101427.00
                                                                                162   101427.10
                                                                                000   101427.20
                                                                                140   101427.30
                                                                                000   101427.40
                                                                                000   101427.50
                                                                                374   101427.60
                                                                                375   101427.70
        %16¤DD%BU,8,8¤,35,47,33,FD,00,00,00,00  @END,CR                         065   101430.00
                                                                                107   101430.10
                                                                                063   101430.20
                                                                                375   101430.30
                                                                                000   101430.40
                                                                                000   101430.50
                                                                                000   101430.60
                                                                                000   101430.70
        DR%BU,64,8¤,3           @DATA RESERVATION                 3,00          101431.00
TYPW4   %16¤DD%BU,8,8¤,FD,31,3B,2D,3D,47,3D,47  @CHAININ                       375   101434.00
                                                                                061   101434.10
                                                                                073   101434.20
                                                                                055   101434.30
                                                                                075   101434.40
                                                                                107   101434.50
                                                                                075   101434.60
                                                                                107   101434.70
        %16¤DD%BU,8,8¤,37,2D,3D,43,35,33,00,00  @FAIL                           067   101435.00
                                                                                055   101435.10
                                                                                075   101435.20
                                                                                103   101435.30
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                                                                                  065   101435.40
                                                                                  063   101435.50
                                                                                  000   101435.60
                                                                                  000   101435.70
        %16¤DD%BU,8,8¤,39,00,53,35,51,53,00,51   @G TEST S                         071   101436.00
                                                                                  000   101436.10
                                                                                  123   101436.20
                                                                                  065   101436.30
                                                                                  121   101436.40
                                                                                  123   101436.50
                                                                                  000   101436.60
                                                                                  121   101436.70
        %16¤DD%BU,8,8¤,37,2D,3D,43,35,33,00,00   @FAIL                            067   101437.00
                                                                                  055   101437.10
                                                                                  075   101437.20
                                                                                  103   101437.30
                                                                                  065   101437.40
                                                                                  063   101437.50
                                                                                  000   101437.60
                                                                                  000   101437.70
        %16¤DD%BU,8,8¤,55,31,31,35,51,51,74,74   @UCCESS..                        125   101440.00
                                                                                  061   101440.10
                                                                                  061   101440.20
                                                                                  065   101440.30
                                                                                  121   101440.40
                                                                                  121   101440.50
                                                                                  164   101440.60
                                                                                  164   101440.70
*
        DR%BU,64,8¤,3            @DATA RESERVATION                  3.00              101441.00
TYPW5   %16¤DD%BU,8,8¤,FD,45,43,53,48,43,35,FE   @CR, MLTPLE, END                 375   101444.00
                                                                                  105   101444.10
                                                                                  103   101444.20
                                                                                  123   101444.30
                                                                                  110   101444.40
                                                                                  103   101444.50
                                                                                  065   101444.60
                                                                                  376   101444.70
*
        DR%BU,64,8¤,3            @DATA RESERVATION                  3.00              101445.00
        %16¤DD%BU,8,8¤,53,35,51,53,00,51,55,31   @TEST SUC                         123   101450.00
                                                                                  065   101450.10
                                                                                  121   101450.20
                                                                                  123   101450.30
                                                                                  000   101450.40
                                                                                  121   101450.50
                                                                                  125   101450.60
                                                                                  061   101450.70
        %16¤DD%BU,8,8¤,31,35,51,51,37,55,43,74   @CESSFUL.                        061   101451.00
                                                                                  065   101451.10
                                                                                  121   101451.20
                                                                                  121   101451.30
                                                                                  067   101451.40
                                                                                  125   101451.50
                                                                                  103   101451.60
                                                                                  164   101451.70
*
        DR%BU,64,8¤,3            @DATA RESERVATION                  3.00              101452.00
TYPW6   %16¤DD%BU,8,8¤,FD,45,31,00,53,51,53,FE   @CR, MC IST, END                 375   101455.00
                                                                                  105   101455.10
                                                                                  061   101455.20
                                                                                  000   101455.30
                                                                                  123   101455.40
                                                                                  121   101455.50
                                                                                  123   101455.60
                                                                                  376   101455.70
```

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                DR%BU,64,8¤,3                                                    3.00                101456.00
                %16¤DD%BU,8,8¤,51,55,31,31,35,51,FE,5F  @SUCCESS,END,Z                    121     101461.00
                                                                                  125     101461.10
                                                                                  061     101461.20
                                                                                  061     101461.30
                                                                                  065     101461.40
                                                                                  121     101461.50
                                                                                  376     101461.60
                                                                                  137     101461.70
                DR%BU,64,8¤,2                                                    2.00     101462.00
                %16¤DD%BU,8,8¤,37,2D,3D,43,35,33,00,00    @FAILEED                067     101464.00
                                                                                  055     101464.10
                                                                                  075     101464.20
                                                                                  103     101464.30
                                                                                  065     101464.40
                                                                                  063     101464.50
                                                                                  000     101464.60
                                                                                  000     101464.70
TYPR1   DR%BU,64,8¤,8           @RESERVED FOR                                   10.00     101465.00
TYPR2   DR%BU,64,8¤,8           @TYPEWRITER                                     10.00     101475.00
TYPR3   DR%BU,64,8¤,4           @READ TESTS                                      4.00     101505.00
TYPR4   DR%BU,64,8¤,8                                                           10.00     101511.00
TYPR5   DR%BU,64,8¤,25                                                          31.00     101521.00
TYPR6   DR%BU,64,8¤,4                                                            4.00     101552.00
TYPR7   DR%BU,64,8¤,10                                                          12.00     101556.00
TYPR8   DR%BU,64,8¤,20                                                          24.00     101570.00
TYPR9   DR%BU,64,8¤,20                                                          24.00     101614.00
TYPR10  DR%BU,64,8¤,20                                                          24.00     101640.00
TYPR11  DR%BU,64,8¤,3                                                            3.00     101664.00
TYPR12  DR%BU,64,8¤,3                                                            3.00     101667.00
TYPR13  DR%BU,64,8¤,4                                                            4.00     101672.00
TYPR14  DR%BU,64,8¤,5                                                            5.00     101676.00
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*
*       TYPEWRITER TESTS
*           TEST ONE-BACKSPACE TEST
*           TEST TWO-RIPPLE TEST
*           TEST THREE-BALL MOVEMENT TEST
*       TEST FOUR - ALL CHARACTER PRINT
*
*
*       BACKSPACE TEST LOOP
*
TWT1    CW%CCR¤,BST1,11,TWT1&1.         @BACKSPACE TEST     102041.00 40 000262.03 C4    101703.00
        CW%CCR¤,BST1&3.,8,TWT1&2.       @TYPES 3 LINES      102044.00 40 000202.03 C5    101704.00
        CW%CDSC¤,BST1&3.,8,TWT1         @LOOP               102044.00 60 000202.03 C3    101705.00
*
*
*       RIPPLE TEST
*
*                                       @RIPPLE 26 LINES
TWT2    CW%CCR¤,RIPL,14,TWT2&1.         @AB...              102054.00 40 000342.03 C7    101706.00
        CW%CCR¤,RIPL0,1,TWT2&2.                             102057.00 40 000022.03 C8    101707.00
        CW%CCR¤,RIPL3,10,TWT2&3.        @BC...              102075.00 40 000242.03 C9    101710.00
        CW%CCR¤,RIPL0,1,TWT2&4.                             102057.00 40 000022.03 CA    101711.00
        CW%CCR¤,RIPL2&2.,3,TWT2&5.      @CD...              102072.00 40 000062.03 CB    101712.00
        CW%CCR¤,RIPL1,7,TWT2&6.                             102060.00 40 000162.03 CC    101713.00
        CW%CCR¤,RIPL0,1,TWT2&7.                             102057.00 40 000022.03 CD    101714.00
        CW%CCR¤,RIPL4&2.,3,TWT2&8.      @DE...              102107.00 40 000062.03 CE    101715.00
        CW%CCR¤,RIPL3,7,TWT2&9.                             102075.00 40 000162.03 CF    101716.00
        CW%CCR¤,RIPL0,1,TWT2A                               102057.00 40 000022.03 D0    101717.00
TWT2A   CW%CCR¤,RIPL1&7.,,6,TWT2A&1.    @EF...              102067.00 40 000142.03 D1    101720.00
        CW%CCR¤,RIPL1,4,TWT2A&2.                            102060.00 40 000102.03 D2    101721.00
        CW%CCR¤,RIPL0,1,TWT2A&3.                            102057.00 40 000022.03 D3    101722.00
        CW%CCR¤,RIPL3&7.,,6,TWT2A&4.    @FG...              102104.00 40 000142.03 D4    101723.00
        CW%CCR¤,RIPL3,4,TWT2A&5.                            102075.00 40 000102.03 D5    101724.00
        CW%CCR¤,RIPL0,1,TWT2A&6.                            102057.00 40 000022.03 D6    101725.00
        CW%CCR¤,RIPL1&4.,,9,TWT2A&7.    @GH...              102064.00 40 000222.03 D7    101726.00
        CW%CCR¤,RIPL1,1,TWT2A&8.                            102060.00 40 000022.03 D8    101727.00
        CW%CCR¤,RIPL0,1,TWT2A&9.                            102057.00 40 000022.03 D9    101730.00
        CW%CCR¤,RIPL3&4.,,9,TWT2B       @HI...              102101.00 40 000222.03 DA    101731.00
TWT2B   CW%CCR¤,RIPL3,1,TWT2B&1.                            102075.00 40 000022.03 DB    101732.00
        CW%CCR¤,RIPL0,1,TWT2B&2.                            102057.00 40 000022.03 DC    101733.00
        CW%CCR¤,RIPL1&1.,,10,TWT2B&3.   @IJ...              102061.00 40 000242.03 DD    101734.00
        CW%CCR¤,RIPL0,1,TWT2B&4.                            102057.00 40 000022.03 DE    101735.00
        CW%CCR¤,RIPL3&1.,,10,TWT2B&5.   @JK...              102076.00 40 000242.03 DF    101736.00
        CW%CCR¤,RIPL0,1,TWT2B&6.                            102057.00 40 000022.03 E0    101737.00
        CW%CCR¤,RIPL2&3.,,2,TWT2B&7.    @KL...              102073.00 40 000042.03 E1    101740.00
        CW%CCR¤,RIPL1,8,TWT2B&8.                            102060.00 40 000202.03 E2    101741.00
        CW%CCR¤,RIPL0,1,TWT2B&9.                            102057.00 40 000022.03 E3    101742.00
        CW%CCR¤,RIPL4&3.,,2,TWT2C       @LM...              102110.00 40 000042.03 E4    101743.00
TWT2C   CW%CCR¤,RIPL3,8,TWT2C&1.                            102075.00 40 000202.03 E5    101744.00
        CW%CCR¤,RIPL0,1,TWT2C&2.                            102057.00 40 000022.03 E6    101745.00
        CW%CCR¤,RIPL2,5,TWT2C&3.        @MN...              102070.00 40 000122.03 E7    101746.00
        CW%CCR¤,RIPL1,5,TWT2C&4.                            102060.00 40 000122.03 E8    101747.00
        CW%CCR¤,RIPL0,1,TWT2C&5.                            102057.00 40 000022.03 E9    101750.00
        CW%CCR¤,RIPL4,5,TWT2C&6.        @NO...              102105.00 40 000122.03 EA    101751.00
        CW%CCR¤,RIPL3,5,TWT2C&7.                            102075.00 40 000122.03 EB    101752.00
        CW%CCR¤,RIPL0,1,TWT2C&8.                            102057.00 40 000022.03 EC    101753.00
        CW%CCR¤,RIPL1&5.,,8,TWT2C&9.    @OP...              102065.00 40 000202.03 ED    101754.00
        CW%CCR¤,RIPL1,2,TWT2D                               102060.00 40 000042.03 EE    101755.00
TWT2D   CW%CCR¤,RIPL0,1,TWT2D&1.                            102057.00 40 000022.03 EF    101756.00
        CW%CCR¤,RIPL3&5.,,8,TWT2D&2.    @PQ...              102102.00 40 000202.03 F0    101757.00
        CW%CCR¤,RIPL3,2,TWT2D&3.                            102075.00 40 000042.03 F1    101760.00
        CW%CCR¤,RIPL0,1,TWT2D&4.                            102057.00 40 000022.03 F2    101761.00
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                CW%CCR¤,RIPL162,,10,TWT2D&5.    @QR...          102062.00 40 000242.03 F3     101762.00
                CW%CCR¤,RIPL0,1,TWT2D&6.                        102057.00 40 000022.03 F4     101763.00
                CW%CCR¤,RIPL362,,10,TWT2D&7.    @RS...          102077.00 40 000242.03 F5     101764.00
                CW%CCR¤,RIPL0,1,TWT2D&8.                        102057.00 40 000022.03 F6     101765.00
                CW%CCR¤,RIPL264,,1,TWT2D&9.     @ST...          102074.00 40 000022.03 F7     101766.00
                CW%CCR¤,RIPL1,9,TWT2E                           102060.00 40 000222.03 F8     101767.00
TWT2E           CW%CCR¤,RIPL0,1,TWT2E&1.                        102057.00 40 000022.03 F9     101770.00
                CW%CCR¤,RIPL464,,1,TWT2E&2.     @TU...          102111.00 40 000022.03 FA     101771.00
                CW%CCR¤,RIPL3,9,TWT2E&3.                        102075.00 40 000222.03 FB     101772.00
                CW%CCR¤,RIPL0,1,TWT2E&4.                        102057.00 40 000022.03 FC     101773.00
                CW%CCR¤,RIPL261,,4,TWT2E&5.     @UV...          102071.00 40 000102.03 FD     101774.00
                CW%CCR¤,RIPL1,6,TWT2E&6.                        102060.00 40 000142.03 FE     101775.00
                CW%CCR¤,RIPL0,1,TWT2E&7.                        102057.00 40 000022.03 FF     101776.00
                CW%CCR¤,RIPL461,,4,TWT2E&8.     @VW...          102106.00 40 000102.04 00     101777.00
                CW%CCR¤,RIPL3,6,TWT2E&9.                        102075.00 40 000142.04 01     102000.00
                CW%CCR¤,RIPL0,1,TWT2F                           102057.00 40 000022.04 02     102001.00
TWT2F           CW%CCR¤,RIPL166,,7,TWT2F&1.     @WX...          102066.00 40 000162.04 03     102002.00
                CW%CCR¤,RIPL1,3,TWT2F&2.                        102060.00 40 000062.04 04     102003.00
                CW%CCR¤,RIPL0,1,TWT2F&3                         102057.00 40 000022.04 05     102004.00
                CW%CCR¤,RIPL366,,7,TWT2F&4.     @XY...          102103.00 40 000162.04 06     102005.00
                CW%CCR¤,RIPL3,3,TWT2F&5.                        102075.00 40 000062.04 07     102006.00
                CW%CCR¤,RIPL0,1,TWT2F&6.                        102057.00 40 000022.04 08     102007.00
                CW%CCR¤,RIPL163,,10,TWT2F&7.    @YZ..           102063.00 40 000242.04 09     102010.00
                CW%CCR¤,RIPL0,1,TWT2F&8.                        102057.00 40 000022.04 0A     102011.00
                CW%CCR¤,RIPL363,,10,TWT2F&9.    @ZA..           102100.00 40 000242.04 0B     102012.00
                CW%CR¤,RIPL5,4,0                                102112.00 00 000100.00 00     102013.00
                @
                @
                @       BALL MOVEMENT TEST LOOP
                @
TWT3            CW%CCR¤,BMT0,15,TWT3&1.         @BALL MOVEMENT TEST
                CW%CCR¤,BMT1,12,TWT3&2.           @PRINTS 10-44 CHAR-   102022.00 40 000362.04 0D     102014.00
                CW%CCR¤,BMT1,12,TWT3&3.         @ACTER LINES AND        102025.00 40 000302.04 0E     102015.00
                CW%CCR¤,BMT1,12,TWT3&4.         @ALL CHARACTERS         102025.00 40 000302.04 0F     102016.00
                CW%CDSC¤,BMT1,12,TWT3           @LOOP                   102025.00 40 000302.04 10     102017.00
                CW%CDSC¤,BMT1,12,TWT3           @LOOP                   102025.00 60 000302.04 0C     102020.00
                @
                @       TEST FOUR
                @
                @       EXECUTE THIS CONTROL WORD FOR AN ALL
                @       CHARACTER PRINT
                @
                @
                CW%CR¤,ALLC,27,0                                102116.00 00 000660.00 00     102021.00
                @
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*
*       TYPEWRITER TEST DATA
*
*
BMT0    DR%BU,64,8¤,3                          @RESERVE 3 LOC.          3.00
BMT1    %16¤DD%BU,8,8¤,FD,70,2F,60,3F,50,4F,40  @CR,8B0JSRK
        %16¤DD%BU,8,8¤,5F,30,6F,20,27,68,37,58  @ZC7  4FW
        %16¤DD%BU,8,8¤,47,48,57,38,67,28,77,71  @NOVG3, 8
        %16¤DD%BU,8,8¤,76,61,66,51,56,41,46,31  @-03SVKNC
        %16¤DD%BU,8,8¤,36,21,26,69,6E,59,5E,49  @F6/47WZO
        %16¤DD%BU,8,8¤,4E,39,3E,29,2E,00,00,FD  @RGJ B   ,CR
        %16¤DD%BU,8,8¤,74,2B,64,3B,54,4B,44,5B  @, 2HUPMX
        %16¤DD%BU,8,8¤,34,6B,24,23,6C,33,5C,43  @E5   6DYL
```

```
102022.00
375 102025.00
160 102025.10
057 102025.20
140 102025.30
077 102025.40
120 102025.50
117 102025.60
100 102025.70
137 102026.00
060 102026.10
157 102026.20
040 102026.30
047 102026.40
150 102026.50
067 102026.60
130 102026.70
107 102027.00
110 102027.10
127 102027.20
070 102027.30
147 102027.40
050 102027.50
167 102027.60
161 102027.70
166 102030.00
141 102030.10
146 102030.20
121 102030.30
126 102030.40
101 102030.50
106 102030.60
061 102030.70
066 102031.00
041 102031.10
046 102031.20
151 102031.30
156 102031.40
131 102031.50
136 102031.60
111 102031.70
116 102032.00
071 102032.10
076 102032.20
051 102032.30
056 102032.40
000 102032.50
000 102032.60
375 102032.70
164 102033.00
053 102033.10
144 102033.20
073 102033.30
124 102033.40
113 102033.50
104 102033.60
133 102033.70
064 102034.00
```

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                                                                                153  102034.10
                                                                                044  102034.20
                                                                                043  102034.30
                                                                                154  102034.40
                                                                                063  102034.50
                                                                                134  102034.60
                                                                                103  102034.70
%16¤DD%BU,8,8¤,4C,53,3C,63,2C,73,72,2D     @QTI1A99A                        114  102035.00
                                                                                123  102035.10
                                                                                074  102035.20
                                                                                143  102035.30
                                                                                054  102035.40
                                                                                163  102035.50
                                                                                162  102035.60
                                                                                055  102035.70
%16¤DD%BU,8,8¤,62,3D,52,4D,42,5D,32,6D     @1ITQLYD6                        142  102036.00
                                                                                075  102036.10
                                                                                122  102036.20
                                                                                115  102036.30
                                                                                102  102036.40
                                                                                135  102036.50
                                                                                062  102036.60
                                                                                155  102036.70
%16¤DD%BU,8,8¤,22,25,6A,35,5A,45,4A,55     @  5EXMPU                        042  102037.00
                                                                                045  102037.10
                                                                                152  102037.20
                                                                                065  102037.30
                                                                                132  102037.40
                                                                                105  102037.50
                                                                                112  102037.60
                                                                                125  102037.70
%16¤DD%BU,8,8¤,3A,65,2A,75,00,00,FC,FC     @H2     BS,BS                    072  102040.00
                                                                                145  102040.10
                                                                                052  102040.20
                                                                                165  102040.30
                                                                                000  102040.40
                                                                                000  102040.50
                                                                                374  102040.60
                                                                                374  102040.70
@
BST1    DR%BU,64,8¤,3                                          3.00              102041.00
        %16¤DD%BU,8,8¤,FD,53,00,00,50,00,3C,50     @T  S IS                    375  102044.00
                                                                                123  102044.10
                                                                                000  102044.20
                                                                                000  102044.30
                                                                                120  102044.40
                                                                                000  102044.50
                                                                                074  102044.60
                                                                                120  102044.70
%16¤DD%BU,8,8¤,FC,FC,FC,FC,FC,FC,3A,3C     @     HI                        374  102045.00
                                                                                374  102045.10
                                                                                374  102045.20
                                                                                374  102045.30
                                                                                374  102045.40
                                                                                374  102045.50
                                                                                072  102045.60
                                                                                074  102045.70
%16¤DD%BU,8,8¤,00,00,00,00,00,2C,00,00     @    A                            000  102046.00
                                                                                000  102046.10
                                                                                000  102046.20
                                                                                000  102046.30
                                                                                000  102046.40
                                                                                054  102046.50
                                                                                000  102046.60
                                                                                000  102046.70
%16¤DD%BU,8,8¤,00,00,00,00,00,00,00,00                                          000  102047.00
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                                                                                000  102047.10
                                                                                000  102047.20
                                                                                000  102047.30
                                                                                000  102047.40
                                                                                000  102047.50
                                                                                000  102047.60
                                                                                000  102047.70
        %16¤DD%BU,8,8¤,00,52,34,50,52,74,FC,FC      @TEST                       000  102050.00
                                                                                122  102050.10
                                                                                064  102050.20
                                                                                120  102050.30
                                                                                122  102050.40
                                                                                164  102050.50
                                                                                374  102050.60
                                                                                374  102050.70
        %16¤DD%BU,8,8¤,FC,FC,FC,FC,FC,FC,FC,FC                                  374  102051.00
                                                                                374  102051.10
                                                                                374  102051.20
                                                                                374  102051.30
                                                                                374  102051.40
                                                                                374  102051.50
                                                                                374  102051.60
                                                                                374  102051.70
        %16¤DD%BU,8,8¤,FC,FC,FC,FC,FC,0F,0D,11      @       BAC                 374  102052.00
                                                                                374  102052.10
                                                                                374  102052.20
                                                                                374  102052.30
                                                                                374  102052.40
                                                                                017  102052.50
                                                                                015  102052.60
                                                                                021  102052.70
        %16¤DD%BU,8,8¤,81,91,8B,0D,11,15,00,00      @KSPACE                     201  102053.00
                                                                                221  102053.10
                                                                                213  102053.20
                                                                                015  102053.30
                                                                                021  102053.40
                                                                                025  102053.50
                                                                                000  102053.60
                                                                                000  102053.70
RIPL    DR%BU,64,8¤,3                                                3,00            102054.00
RIPL0   %16¤DD%BU,8,8¤,FC,00,00,FC,FC,00,00,FD      @CARR-RET                   374  102057.00
                                                                                000  102057.10
                                                                                000  102057.20
                                                                                374  102057.30
                                                                                374  102057.40
                                                                                000  102057.50
                                                                                000  102057.60
                                                                                375  102057.70
RIPL1   %16¤DD%BU,8,8¤,2D,2F,31,33,35,37,39,3B      @ABCDEFGH                   055  102060.00
                                                                                057  102060.10
                                                                                061  102060.20
                                                                                063  102060.30
                                                                                065  102060.40
                                                                                067  102060.50
                                                                                071  102060.60
                                                                                073  102060.70
        %16¤DD%BU,8,8¤,3D,3F,41,43,45,47,49,4B      @IJKLMNOP                   075  102061.00
                                                                                077  102061.10
                                                                                101  102061.20
                                                                                103  102061.30
                                                                                105  102061.40
                                                                                107  102061.50
                                                                                111  102061.60
                                                                                113  102061.70
        %16¤DD%BU,8,8¤,4D,4F,51,53,55,57,59,5B      @QRSTUVWX                   115  102062.00
                                                                                117  102062.10
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                                                                                121     102062.20
                                                                                123     102062.30
                                                                                125     102062.40
                                                                                127     102062.50
                                                                                131     102062.60
                                                                                133     102062.70
        %16¤DD%BU,8,8¤,5D,5F,2D,2F,31,33,35,37      @YZABCDEF                   135     102063.00
                                                                                137     102063.10
                                                                                055     102063.20
                                                                                057     102063.30
                                                                                061     102063.40
                                                                                063     102063.50
                                                                                065     102063.60
                                                                                067     102063.70
        %16¤DD%BU,8,8¤,39,3B,3D,3F,41,43,45,47      @GHIJKLMN                   071     102064.00
                                                                                073     102064.10
                                                                                075     102064.20
                                                                                077     102064.30
                                                                                101     102064.40
                                                                                103     102064.50
                                                                                105     102064.60
                                                                                107     102064.70
        %16¤DD%BU,8,8¤,49,4B,4D,4F,51,53,55,57      @OPQRSTUV                   111     102065.00
                                                                                113     102065.10
                                                                                115     102065.20
                                                                                117     102065.30
                                                                                121     102065.40
                                                                                123     102065.50
                                                                                125     102065.60
                                                                                127     102065.70
        %16¤DD%BU,8,8¤,59,5B,5D,5F,2D,2F,31,33      @WXYZABCD                   131     102066.00
                                                                                133     102066.10
                                                                                135     102066.20
                                                                                137     102066.30
                                                                                055     102066.40
                                                                                057     102066.50
                                                                                061     102066.60
                                                                                063     102066.70
            %16¤DD%BU,8,8¤,35,37,39,3B,3D,3F,41,43  @EFGHIJKL                   065     102067.00
                                                                                067     102067.10
                                                                                071     102067.20
                                                                                073     102067.30
                                                                                075     102067.40
                                                                                077     102067.50
                                                                                101     102067.60
                                                                                103     102067.70
    RIPL2   %16¤DD%BU,8,8¤,45,47,49,4B,4D,4F,51,53  @MNOPQRST                   105     102070.00
                                                                                107     102070.10
                                                                                111     102070.20
                                                                                113     102070.30
                                                                                115     102070.40
                                                                                117     102070.50
                                                                                121     102070.60
                                                                                123     102070.70
            %16¤DD%BU,8,8¤,55,57,59,5B,5D,5F,2D,2F  @UVWXYZAB                   125     102071.00
                                                                                127     102071.10
                                                                                131     102071.20
                                                                                133     102071.30
                                                                                135     102071.40
                                                                                137     102071.50
                                                                                055     102071.60
                                                                                057     102071.70
            %16¤DD%BU,8,8¤,31,33,35,37,39,3B,3D,3F  @CDEFGHIJ                   061     102072.00
                                                                                063     102072.10
                                                                                065     102072.20
                                                                                067     102072.30
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                                                                                071   102072.40
                                                                                073   102072.50
                                                                                075   102072.60
                                                                                077   102072.70
            %16¤DD%BU,8,8¤,41,43,45,47,49,4B,4D,4F      @KLMNOPQR                 101   102073.00
                                                                                103   102073.10
                                                                                105   102073.20
                                                                                107   102073.30
                                                                                111   102073.40
                                                                                113   102073.50
                                                                                115   102073.60
                                                                                117   102073.70
            %16¤DD%BU,8,8¤,51,53,55,57,59,5B,5D,5F      @STUVWXYZ                 121   102074.00
                                                                                123   102074.10
                                                                                125   102074.20
                                                                                127   102074.30
                                                                                131   102074.40
                                                                                133   102074.50
                                                                                135   102074.60
                                                                                137   102074.70
            @
RIPL3       %16¤DD%BU,8,8¤,2F,31,33,35,37,39,3B,3D      @BCDEFGHI                 057   102075.00
                                                                                061   102075.10
                                                                                063   102075.20
                                                                                065   102075.30
                                                                                067   102075.40
                                                                                071   102075.50
                                                                                073   102075.60
                                                                                075   102075.70
            %16¤DD%BU,8,8¤,3F,41,43,45,47,49,4B,4D      @JKLMNOPQ                 077   102076.00
                                                                                101   102076.10
                                                                                103   102076.20
                                                                                105   102076.30
                                                                                107   102076.40
                                                                                111   102076.50
                                                                                113   102076.60
                                                                                115   102076.70
            %16¤DD%BU,8,8¤,4F,51,53,55,57,59,5B,5D      @RSTUVWXY                 117   102077.00
                                                                                121   102077.10
                                                                                123   102077.20
                                                                                125   102077.30
                                                                                127   102077.40
                                                                                131   102077.50
                                                                                133   102077.60
                                                                                135   102077.70
            %16¤DD%BU,8,8¤,5F,2D,2F,31,33,35,37,39      @ZABCDEFG                 137   102100.00
                                                                                055   102100.10
                                                                                057   102100.20
                                                                                061   102100.30
                                                                                063   102100.40
                                                                                065   102100.50
                                                                                067   102100.60
                                                                                071   102100.70
            %16¤DD%BU,8,8¤,3B,3D,3F,41,43,45,47,49      @HIJKLMNO                 073   102101.00
                                                                                075   102101.10
                                                                                077   102101.20
                                                                                101   102101.30
                                                                                103   102101.40
                                                                                105   102101.50
                                                                                107   102101.60
                                                                                111   102101.70
            %16¤DD%BU,8,8¤,4B,4D,4F,51,53,55,57,59      @PQRSTUVW                 113   102102.00
                                                                                115   102102.10
                                                                                117   102102.20
                                                                                121   102102.30
                                                                                123   102102.40
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        %16¤DD%BU,8,8¤,5B,5D,5F,2D,2F,31,33,35     @XYZABCDE
        %16¤DD%BU,8,8¤,37,39,3B,3D,3F,41,43,45     @FGHIJKLM
        %16¤DD%BU,8,8¤,47,49,4B,4D,4F,51,53,55     @NOPQRSTU
RIPL4   %16¤DD%BU,8,8¤,57,59,5B,5D,5F,2D,2F,31     @VWXYZABC
        %16¤DD%BU,8,8¤,33,35,37,39,3B,3D,3F,41     @DEFGHIJK
        %16¤DD%BU,8,8¤,43,45,47,49,4B,4D,4F,51     @LMNOPQRS
        %16¤DD%BU,8,8¤,53,55,57,59,5B,5D,5F,2D     @TUVWXYZA
RIPL5   %16¤DD%BU,8,8¤,FD,53,3B,3D,51,00,3D,51     @CR, THIS IS
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```
125 102102.50
127 102102.60
131 102102.70
133 102103.00
135 102103.10
137 102103.20
055 102103.30
057 102103.40
061 102103.50
063 102103.60
065 102103.70
067 102104.00
071 102104.10
073 102104.20
075 102104.30
077 102104.40
101 102104.50
103 102104.60
105 102104.70
107 102105.00
111 102105.10
113 102105.20
115 102105.30
117 102105.40
121 102105.50
123 102105.60
125 102105.70
127 102106.00
131 102106.10
133 102106.20
135 102106.30
137 102106.40
055 102106.50
057 102106.60
061 102106.70
063 102107.00
065 102107.10
067 102107.20
071 102107.30
073 102107.40
075 102107.50
077 102107.60
101 102107.70
103 102110.00
105 102110.10
107 102110.20
111 102110.30
113 102110.40
115 102110.50
117 102110.60
121 102110.70
123 102111.00
125 102111.10
127 102111.20
131 102111.30
133 102111.40
135 102111.50
137 102111.60
055 102111.70
375 102112.00
123 102112.10
073 102112.20
075 102112.30
121 102112.40
000 102112.50
075 102112.60
```

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                %16¤DD%BU,8,8¤,53,3B,35,00,35,47,33,00     @THE END                                                  121  102112.70
                                                                                                                    123  102113.00
                                                                                                                    073  102113.10
                                                                                                                    065  102113.20
                                                                                                                    000  102113.30
                                                                                                                    065  102113.40
                                                                                                                    107  102113.50
                                                                                                                    063  102113.60
                                                                                                                    000  102113.70
                %16¤DD%BU,8,8¤,49,37,00,4F,3D,4B,4B,43     @OF RIPPL                                                 111  102114.00
                                                                                                                    067  102114.10
                                                                                                                    000  102114.20
                                                                                                                    117  102114.30
                                                                                                                    075  102114.40
                                                                                                                    113  102114.50
                                                                                                                    113  102114.60
                                                                                                                    103  102114.70
                %16¤DD%BU,8,8¤,35,00,53,35,51,53,74,74     @E TEST..                                                 065  102115.00
                                                                                                                    000  102115.10
                                                                                                                    123  102115.20
                                                                                                                    065  102115.30
                                                                                                                    121  102115.40
                                                                                                                    123  102115.50
                                                                                                                    164  102115.60
                                                                                                                    164  102115.70
*
                CNOP
*
*               RED ALPHABET
*
ALLC            DR%BU,64,8¤,%3¤                                                    3.00                             102116.00
                %16¤DD%BU,8,8¤,FD,0C,0D,0E,0F,10,11,12                                                          375  102121.00
                                                                                                                014  102121.10
                                                                                                                015  102121.20
                                                                                                                016  102121.30
                                                                                                                017  102121.40
                                                                                                                020  102121.50
                                                                                                                021  102121.60
                                                                                                                022  102121.70
                %16¤DD%BU,8,8¤,13,14,15,16,17,18,19,1A                                                          023  102122.00
                                                                                                                024  102122.10
                                                                                                                025  102122.20
                                                                                                                026  102122.30
                                                                                                                027  102122.40
                                                                                                                030  102122.50
                                                                                                                031  102122.60
                                                                                                                032  102122.70
                %16¤DD%BU,8,8¤,1B,1C,1D,1E,1F,80,81,82                                                          033  102123.00
                                                                                                                034  102123.10
                                                                                                                035  102123.20
                                                                                                                036  102123.30
                                                                                                                037  102123.40
                                                                                                                200  102123.50
                                                                                                                201  102123.60
                                                                                                                202  102123.70
                %16¤DD%BU,8,8¤,83,84,85,86,87,88,89,8A                                                          203  102124.00
                                                                                                                204  102124.10
                                                                                                                205  102124.20
                                                                                                                206  102124.30
                                                                                                                207  102124.40
                                                                                                                210  102124.50
                                                                                                                211  102124.60
                                                                                                                212  102124.70
                %16¤DD%BU,8,8¤,8B,8C,8D,8E,8F,90,91,92                                                          213  102125.00
                                                                                                                214  102125.10
                                                                                                                215  102125.20
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                                                                    216  102125.30
                                                                    217  102125.40
                                                                    220  102125.50
                                                                    221  102125.60
                                                                    222  102125.70
%16¤DD%BU,8,8¤,93,94,95,96,97,98,99,9A                              223  102126.00
                                                                    224  102126.10
                                                                    225  102126.20
                                                                    226  102126.30
                                                                    227  102126.40
                                                                    230  102126.50
                                                                    231  102126.60
                                                                    232  102126.70
%16¤DD%BU,8,8¤,9B,9C,9D,9E,9F,00,00,00                              233  102127.00
                                                                    234  102127.10
                                                                    235  102127.20
                                                                    236  102127.30
                                                                    237  102127.40
                                                                    000  102127.50
                                                                    000  102127.60
                                                                    000  102127.70

BLACK ALPHABET

%16¤DD%BU,8,8¤,FD,2C,2D,2E,2F,30,31,32                              375  102130.00
                                                                    054  102130.10
                                                                    055  102130.20
                                                                    056  102130.30
                                                                    057  102130.40
                                                                    060  102130.50
                                                                    061  102130.60
                                                                    062  102130.70
%16¤DD%BU,8,8¤,33,34,35,36,37,38,39,3A                              063  102131.00
                                                                    064  102131.10
                                                                    065  102131.20
                                                                    066  102131.30
                                                                    067  102131.40
                                                                    070  102131.50
                                                                    071  102131.60
                                                                    072  102131.70
%16¤DD%BU,8,8¤,3B,3C,3D,3E,3F,40,41,42                              073  102132.00
                                                                    074  102132.10
                                                                    075  102132.20
                                                                    076  102132.30
                                                                    077  102132.40
                                                                    100  102132.50
                                                                    101  102132.60
                                                                    102  102132.70
%16¤DD%BU,8,8¤,43,44,45,46,47,48,49,4A                              103  102133.00
                                                                    104  102133.10
                                                                    105  102133.20
                                                                    106  102133.30
                                                                    107  102133.40
                                                                    110  102133.50
                                                                    111  102133.60
                                                                    112  102133.70
%16¤DD%BU,8,8¤,4B,4C,4D,4E,4F,50,51,52                              113  102134.00
                                                                    114  102134.10
                                                                    115  102134.20
                                                                    116  102134.30
                                                                    117  102134.40
                                                                    120  102134.50
                                                                    121  102134.60
                                                                    122  102134.70
%16¤DD%BU,8,8¤,53,54,55,56,57,58,59,5A                              123  102135.00
                                                                    124  102135.10
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                                                                    125   102135.20
                                                                    126   102135.30
                                                                    127   102135.40
                                                                    130   102135.50
                                                                    131   102135.60
                                                                    132   102135.70
%16¤DD%BU,8,8¤,5B,5C,5D,5E,5F,00,00,00                              133   102136.00
                                                                    134   102136.10
                                                                    135   102136.20
                                                                    136   102136.30
                                                                    137   102136.40
                                                                    000   102136.50
                                                                    000   102136.60
                                                                    000   102136.70

        RED NUMBERS & SPECIALS

%16¤DD%BU,8,8¤,FD,01,02,03,04,05,06,07                              375   102137.00
                                                                    001   102137.10
                                                                    002   102137.20
                                                                    003   102137.30
                                                                    004   102137.40
                                                                    005   102137.50
                                                                    006   102137.60
                                                                    007   102137.70
%16¤DD%BU,8,8¤,08,09,0A,0B,A0,A1,A2,A3                              010   102140.00
                                                                    011   102140.10
                                                                    012   102140.20
                                                                    013   102140.30
                                                                    240   102140.40
                                                                    241   102140.50
                                                                    242   102140.60
                                                                    243   102140.70
%16¤DD%BU,8,8¤,A4,A5,A6,A7,A8,A9,AA,AB                              244   102141.00
                                                                    245   102141.10
                                                                    246   102141.20
                                                                    247   102141.30
                                                                    250   102141.40
                                                                    251   102141.50
                                                                    252   102141.60
                                                                    253   102141.70
%16¤DD%BU,8,8¤,AC,AD,AE,AF,B0,B1,B2,B3                              254   102142.00
                                                                    255   102142.10
                                                                    256   102142.20
                                                                    257   102142.30
                                                                    260   102142.40
                                                                    261   102142.50
                                                                    262   102142.60
                                                                    263   102142.70
%16¤DD%BU,8,8¤,B4,B5,B6,B7,00,00,00,00                              264   102143.00
                                                                    265   102143.10
                                                                    266   102143.20
                                                                    267   102143.30
                                                                    000   102143.40
                                                                    000   102143.50
                                                                    000   102143.60
                                                                    000   102143.70

        BLACK NUMBERS & SPECIALS

%16¤DD%BU,8,8¤,FD,20,21,22,23,24,25,26                              375   102144.00
                                                                    040   102144.10
                                                                    041   102144.20
                                                                    042   102144.30
                                                                    043   102144.40
                                                                    044   102144.50
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                                                                    045   102144.60
                                                                    046   102144.70
%16¤DD%BU,8,8¤,27,28,29,2A,2B,60,61,62                              047   102145.00
                                                                    050   102145.10
                                                                    051   102145.20
                                                                    052   102145.30
                                                                    053   102145.40
                                                                    140   102145.50
                                                                    141   102145.60
                                                                    142   102145.70
%16¤DD%BU,8,8¤,63,64,65,66,67,68,69,6A                              143   102146.00
                                                                    144   102146.10
                                                                    145   102146.20
                                                                    146   102146.30
                                                                    147   102146.40
                                                                    150   102146.50
                                                                    151   102146.60
                                                                    152   102146.70
%16¤DD%BU,8,8¤,6B,6C,6D,6E,6F,70,71,72                              153   102147.00
                                                                    154   102147.10
                                                                    155   102147.20
                                                                    156   102147.30
                                                                    157   102147.40
                                                                    160   102147.50
                                                                    161   102147.60
                                                                    162   102147.70
%16¤DD%BU,8,8¤,73,74,75,76,77,00,00,00                              163   102150.00
                                                                    164   102150.10
                                                                    165   102150.20
                                                                    166   102150.30
                                                                    167   102150.40
                                                                    000   102150.50
                                                                    000   102150.60
                                                                    000   102150.70
CNOP
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*       CARD PUNCH TEST
*
*           TEST ONE-NON-ECC MODE
*           TEST TWO-ECC MODE
*
*       TEST ONE-NON-ECC MODE-15 WORDS PER CARD
*           TABLE OF STARTING POSITION OF WORDS PUNCHED
*
*            WORD     COLUMN     ROW      WORD     COLUMN     ROW
*              1          1       12         2          6       2
*              3         11        6         4         17      12
*              5         22        2         6         27       6
*              7         33       12         8         38       2
*              9         43        6        10         49      12
*             11         55        2        12         59       6
*             13         65       12        14         70       2
*             15         75        6
*
PCH1    CW%CR¤,PWD1,15,0                @PUNCH ONE CARD          102450.00 00 000360.00 00     102151.00
*
*
*       THE FOLLOWING CWS PUNCH 13 CARDS DIAGONAL PATTERN
*
        CW%CCR¤,PWD1,13,$&1.                                     102450.00 40 000322.04 6B     102152.00
        CW%CDSC¤,PWD1,2,$&1.            @PATTERN TEST-PUNCH 13 CARDS 102450.00 60 000042.04 6C 102153.00
        CW%CCR¤,PWD1&1.,12,$&1.         @TOTAL OF 13 CARDS.      102451.00 40 000302.04 6D     102154.00
        CW%CDSC¤,PWD1,3,$&1.                                     102450.00 60 000062.04 6E     102155.00
        CW%CCR¤,PWD1&2.,11,$&1.         @CARD 3                  102452.00 40 000262.04 6F     102156.00
        CW%CDSC¤,PWD1,4,$&1.                                     102450.00 60 000102.04 70     102157.00
        CW%CCR¤,PWD1&3.,10,$&1.         @CARD 4                  102453.00 40 000242.04 71     102160.00
        CW%CDSC¤,PWD1,5,$&1.                                     102450.00 60 000122.04 72     102161.00
        CW%CCR¤,PWD1&4.,9,$&1.          @CARD 5                  102454.00 40 000222.04 73     102162.00
        CW%CDSC¤,PWD1,6,$&1.                                     102450.00 60 000142.04 74     102163.00
        CW%CCR¤,PWD1&5.,8,$&1.          @CARD 6                  102455.00 40 000202.04 75     102164.00
        CW%CDSC¤,PWD1,7,$&1.                                     102450.00 60 000162.04 76     102165.00
        CW%CCR¤,PWD1&6.,7,$&1.          @CARD 7                  102456.00 40 000162.04 77     102166.00
        CW%CDSC¤,PWD1,8,$&1.                                     102450.00 60 000202.04 78     102167.00
        CW%CCR¤,PWD1&7.,6,$&1.          @CARD 8                  102457.00 40 000142.04 79     102170.00
        CW%CDSC¤,PWD1,9,$&1.                                     102450.00 60 000222.04 7A     102171.00
        CW%CCR¤,PWD1&8.,5,$&1.          @CARD 9                  102460.00 40 000122.04 7B     102172.00
        CW%CDSC¤,PWD1,10,$&1.                                    102450.00 60 000242.04 7C     102173.00
        CW%CCR¤,PWD1&9.,4,$&1.          @CARD 10                 102461.00 40 000102.04 7D     102174.00
        CW%CDSC¤,PWD1,11,$&1.                                    102450.00 60 000262.04 7E     102175.00
        CW%CCR¤,PWD1&10.,3,$&1.         @CARD 11                 102462.00 40 000062.04 7F     102176.00
        CW%CDSC¤,PWD1,12,$&1.                                    102450.00 60 000302.04 80     102177.00
        CW%CCR¤,PWD1&11.,2,$&1.         @CARD 12                 102463.00 40 000042.04 81     102200.00
        CW%CDSC¤,PWD1,13,$&1.                                    102450.00 60 000322.04 82     102201.00
        CW%CCR¤,PWD1&12.,1,$&1.                                  102464.00 40 000022.04 83     102202.00
        CW%CCR¤,PWD1,13,$&1.                                     102450.00 40 000322.04 84     102203.00
        CW%CD¤,PWD1,1,0                                          102450.00 20 000020.00 00     102204.00
*
```

TEST TWO-ECC MODE-13 WORDS PER CARD

TABLE OF STARTING POSITION OF WORDS PUNCHED
ALL WORDS BEGIN IN ROW 12.

WORD	COLUMN	WORD	COLUMN
1	1	2	7
3	13	4	19
5	25	6	31
7	37	8	43
9	49	10	55
11	61	12	67
13	73		

TABLE OF BITS ON WHICH ECC BIT IS BASED

ECC BITS	DATA BITS
C-0	0-32
C-1	1, 3, 5,...61, 63 → 32
C-2	2-3, 6-7, 10-11, ... 62-63 → 16
C-4	4-7, 12-15, ... 60-63 → 8
C-8	8-15, 24-31, 40-47, 56-63 → 4
C-16	16-31 ,48-63 → 2
C-32	0, 32-63 → 1

C-T IS BASED ON OVERALL PARITY INCLUDING ECC BITS

SET PUNCH TO ECC MODE. CONTROL CODE 00101111

```
PCH2    CW%CR¤,PWD2,13,0          @PUNCH 1 ECC-CARD          102465.00 00 000320.00 00     102205.00
```

THE FOLLOWING CWS PUNCH 9 CARDS ECC MODE

FLOATING ONE C-BIT PATTERN

PUNCH NINE CARDS

CARD	FIRST WORD C-BITS
@ 1	200
@ 2	010
@ 3	000
@ 4	020
@ 5	001
@ 6	040
@ 7	002
@ 8	100
@ 9	004

```
        CW%CCR¤,PWD2,9,$&1.                          102465.00 40 000222.04 87     102206.00
        CW%CDSC¤,PWD2,4,$&1.                         102465.00 60 000102.04 88     102207.00
        CW%CCR¤,PWD2&4.,5,$&1.                       102471.00 40 000122.04 89     102210.00
        CW%CDSC¤,PWD2,8,$&1.                         102465.00 60 000202.04 8A     102211.00
        CW%CCR¤,PWD2&8.,1,$&1.                       102475.00 40 000022.04 8B     102212.00
        CW%CCR¤,PWD2,9,$&1.                          102465.00 40 000222.04 8C     102213.00
        CW%CDSC¤,PWD2,3,$&1.                         102465.00 60 000062.04 8D     102214.00
        CW%CCR¤,PWD2&3.,6,$&1.                       102470.00 40 000142.04 8E     102215.00
        CW%CDSC¤,PWD2,7,$&1.                         102465.00 60 000162.04 8F     102216.00
        CW%CCR¤,PWD2&7.,2,$&1.                       102474.00 40 000042.04 90     102217.00
        CW%CCR¤,PWD2,9,$&1.                          102465.00 40 000222.04 91     102220.00
```

```
CW%CDSC¤,PWD2,2,$&1.                102465.00 60 000042.04 92      102221.00
CW%CCR¤,PWD2&2.,7,$&1.              102467.00 40 000162.04 93      102222.00
CW%CDSC¤,PWD2,6,$&1.                102465.00 60 000142.04 94      102223.00
CW%CCR¤,PWD2&6.,3,$&1.              102473.00 40 000062.04 95      102224.00
CW%CCR¤,PWD2,9,$&1.                 102465.00 40 000222.04 96      102225.00
CW%CDSC¤,PWD2,1,$&1.                102465.00 60 000022.04 97      102226.00
CW%CCR¤,PWD2&1.,8,$&1.               102466.00 40 000202.04 98      102227.00
CW%CDSC¤,PWD2,5,$&1.                102465.00 60 000122.04 99      102230.00
CW%CCR¤,PWD2&5.,4,$&1.               102472.00 40 000102.04 9A      102231.00
CW%CD¤,PWD2,9,$&1.                  102465.00 20 000222.04 9B      102232.00
```

THE FOLLOWING CWS PUNCH 9 CARDS ECC MODE

FLOATING ZERO C-BIT PATTERN

PUNCH NINE CARDS	CARD	FIRST WORD C-BITS
	1	377
	2	357
	3	376
	4	337
	5	375
	6	277
	7	373
	8	177
	9	367

SET PUNCH TO ECC MODE. CONTROL CODE 00101111

```
CW%CCR¤,PWD3,9,$&1.                    102502.00 40 000222.04 9C    102233.00
CW%CDSC¤,PWD3,4,$&1.                   102502.00 60 000102.04 9D    102234.00
CW%CCR¤,PWD3&4.,5,$&1.                 102506.00 40 000122.04 9E    102235.00
CW%CDSC¤,PWD3,8,$&1.                   102502.00 60 000202.04 9F    102236.00
CW%CCR¤,PWD3&8.,1,$&1.                 102512.00 40 000022.04 A0    102237.00
CW%CCR¤,PWD3,9,$&1.                    102502.00 40 000222.04 A1    102240.00
CW%CDSC¤,PWD3,3,$&1.                   102502.00 60 000062.04 A2    102241.00
CW%CCR¤,PWD3&3.,6,$&1.                 102505.00 40 000142.04 A3    102242.00
CW%CDSC¤,PWD3,7,$&1.                   102502.00 60 000162.04 A4    102243.00
CW%CCR¤,PWD2&3.,2,$&1.                 102470.00 40 000042.04 A5    102244.00
CW%CCR¤,PWD3,9,$&1.                    102502.00 40 000222.04 A6    102245.00
CW%CDSC¤,PWD3,2,$&1.                   102502.00 60 000042.04 A7    102246.00
CW%CCR¤,PWD3&2.,7,$&1.                 102504.00 40 000162.04 A8    102247.00
CW%CDSC¤,PWD3,6,$&1.                   102502.00 60 000142.04 A9    102250.00
CW%CCR¤,PWD3&6.,3,$&1.                 102510.00 40 000062.04 AA    102251.00
CW%CCR¤,PWD3,9,$&1.                    102502.00 40 000222.04 AB    102252.00
CW%CDSC¤,PWD3,1,$&1.                   102502.00 60 000022.04 AC    102253.00
CW%CCR¤,PWD3&1.,8,$&1.                 102503.00 40 000202.04 AD    102254.00
CW%CDSC¤,PWD3,5,$&1.                   102502.00 60 000122.04 AE    102255.00
CW%CCR¤,PWD3&5.,4,$&1.                 102507.00 40 000102.04 AF    102256.00
CW%CD¤,PWD3,9,0                        102502.00 20 000220.00 00    102257.00
```

```
*
*       PUNCH TEST USING IOS DATA
*
*       TO CHECK CARDS PUNCHED, A PRINTOUT OF READ
*       IN AREA AND WRITE AREA IS PROVIDED. WRITE
*       AREA WORDS ARE PRINTED FIRST.
*                                   @NON-ECC MODE.
PCH3    CW%CD¤,PWD4,30,0            @PUNCH 2 CARDS NON-ECC MODE   102513.00 20 000740.00 00   102260.00
        CW%CD¤,PRES3,30,0           @READ 2 CARDS NON-ECC MODE    102715.00 20 000740.00 00   102261.00
        CW%CDSC¤,PRES1,7,$&1.       @IDENTIFICATION               102677.00 60 000162.04 B3   102262.00
        CW%CDSC¤,PWD4,15,$&1.                                     102513.00 60 000362.04 B4   102263.00
        CW%CDSC¤,PWD4&15.,15,$&1.                                 102532.00 60 000362.04 B5   102264.00
        CW%CDSC¤,PRES2,7,$&1.       @IDENTIFICATION WORD          102706.00 60 000162.04 B6   102265.00
        CW%CDSC¤,PRES3,15,$&1.                                    102715.00 60 000362.04 B7   102266.00
        CW%CR¤,PRES3&15.,15,0                                     102734.00 00 000360.00 00   102267.00
*
*
*       SET PUNCH AND READER TO ECC MODE. CONTROL CODE  00101111
*
PCH4    CW%CD¤,PWD5,26,0            @PUNCH 2 CARDS ECC MODE       102551.00 20 000640.00 00   102270.00
        CW%CD¤,PRES3A,26,0          @READ 2 CARDS ECC MODE        102753.00 20 000640.00 00   102271.00
        CW%CDSC¤,PRES1,7,$&1.       @IDENTIFICATION               102677.00 60 000162.04 BB   102272.00
        CW%CDSC¤,PWD5,13,$&1.                                     102551.00 60 000322.04 BC   102273.00
        CW%CDSC¤,PWD5&13.,$&1.                                    102566.00 62 045720.00 00   102274.00
        CW%CDSC¤,PRES2,7,$&1.       @IDENTIFICATION WORD          102706.00 60 000162.04 BE   102275.00
        CW%CDSC¤,PRES3A,13,$&1.                                   102753.00 60 000322.04 BF   102276.00
        CW%CR¤,PRES3A&13.,$&1.                                    102770.00 02 046000.00 00   102277.00
*
```

```
@       EXTENDED PUNCH TEST
@         CONTROL WORDS ARE PROVIDED FOR ECC OR NON-ECC MODE
@         FOR CHECKING, THE FOLLOWING IS INCLUDED-
@           1.CONTROL WORDS FOR READING PUNCH TEST OUTPUT.
@             ...CARDS MUST BE READ IN SAME MODE AS PUNCHED.
@           2.CONTROL WORDS TO PRINT OUT CORRECT DATA
@             AND TEST DATA,EACH IDENTIFIED, CORRECT DATA
@             WILL BE PRINTED FIRST..
@                                 @NON-ECC MODE-CF-1, 10 CARDS
@
PCH5    CW%CDSC¤,PWD6,15,$&1.     @CARD 6       102603.00 60 000362.04 C1    102300.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 C2    102301.00
        CW%CDSC¤,PWD6C,5,$&1.     @CARD 7       102622.00 60 000122.04 C3    102302.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 C4    102303.00
        CW%CDSC¤,PWD6D,5,$&1.     @CARD 10      102627.00 60 000122.04 C5    102304.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 C6    102305.00
        CW%CDSC¤,PWD6E,5,$&1.     @CARD 11      102634.00 60 000122.04 C7    102306.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 C8    102307.00
        CW%CDSC¤,PWD6F,5,$&1.     @CARD 12      102641.00 60 000122.04 C9    102310.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 CA    102311.00
        CW%CDSC¤,PWD6G,5,$&1.     @CARD 13      102646.00 60 000122.04 CB    102312.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 CC    102313.00
        CW%CDSC¤,PWD6H,5,$&1.     @CARD 14      102653.00 60 000122.04 CD    102314.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 CE    102315.00
        CW%CDSC¤,PWD6J,5,$&1.     @CARD 15      102660.00 60 000122.04 CF    102316.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 D0    102317.00
        CW%CDSC¤,PWD6K,5,$&1.     @CARD 16      102665.00 60 000122.04 D1    102320.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 D2    102321.00
        CW%CD¤,PWD6L,5,0          @CARD 17      102672.00 20 000120.00 00    102322.00
@
        CW%CD¤,PRES2,150,0        @USE THIS CW TO READ                       102323.00
                                  @CARDS
@
@       USE THE FOLLOWING CONTROL WORDS FOR PRINTOUT
@
        CW%CDSC¤,PRES1,7,$&1.     @IDENTIFICATION   102677.00 60 000162.04 D5    102324.00
@
        CW%CDSC¤,PWD6,15,$&1.                   102603.00 60 000362.04 D6    102325.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 D7    102326.00
        CW%CDSC¤,PWD6C,5,$&1.                   102622.00 60 000122.04 D8    102327.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 D9    102330.00
        CW%CDSC¤,PWD6D,5,$&1.                   102627.00 60 000122.04 DA    102331.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 DB    102332.00
        CW%CDSC¤,PWD6E,5,$&1.                   102634.00 60 000122.04 DC    102333.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 DD    102334.00
        CW%CDSC¤,PWD6F,5,$&1.                   102641.00 60 000122.04 DE    102335.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 DF    102336.00
        CW%CDSC¤,PWD6G,5,$&1.                   102646.00 60 000122.04 E0    102337.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 E1    102340.00
        CW%CDSC¤,PWD6H,5,$&1.                   102653.00 60 000122.04 E2    102341.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 E3    102342.00
        CW%CDSC¤,PWD6J,5,$&1.                   102660.00 60 000122.04 E4    102343.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 E5    102344.00
        CW%CDSC¤,PWD6K,5,$&1.                   102665.00 60 000122.04 E6    102345.00
        CW%CCR¤,PWD6,10,$&1.                    102603.00 40 000242.04 E7    102346.00
        CW%CDSC¤,PWD6L,5,$&1.                   102672.00 60 000122.04 E8    102347.00
@
        CW%CDSC¤,PRES2,7,$&1.     @IDENTIFICATION WORD 102706.00 60 000162.04 E9    102350.00
        CW%CDSC¤,PRES4,15,$&1.    @FROM READ AREA      103005.00 60 000362.04 EA    102351.00
        CW%CDSC¤,PRES5,15,$&1.                         103024.00 60 000362.04 EB    102352.00
        CW%CDSC¤,PRES6,15,$&1.                         103043.00 60 000362.04 EC    102353.00
        CW%CDSC¤,PRES7,15,$&1.                         103062.00 60 000362.04 ED    102354.00
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```
CW%CDSC¤,PRES8,15,$&1.          103101.00 60 000362.04 EE      102355.00
CW%CDSC¤,PRES9,15,$&1.          103120.00 60 000362.04 EF      102356.00
CW%CDSC¤,PRES10,15,$&1.         103137.00 60 000362.04 F0      102357.00
CW%CDSC¤,PRES11,15,$&1.         103156.00 60 000362.04 F1      102360.00
CW%CDSC¤,PRES12,15,$&1.         103175.00 60 000362.04 F2      102361.00
CW%CR¤,PRES13,15,0              103214.00 00 000360.00 00      102362.00
```

```
*
*       SET PUNCH AND READER TO ECC MODE. CONTROL CODE  00101111
*
*
*
*                                   @ECC MODE-CF-1,10 CARDS
*
PCH6    CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.04 F4     102363.00
        CW%CDSC¤,PWD6B,5,$61.       @CARD 6                     102615.00 60 000122.04 F5     102364.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.04 F6     102365.00
        CW%CDSC¤,PWD6C,5,$61.       @CARD 7                     102622.00 60 000122.04 F7     102366.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.04 F8     102367.00
        CW%CDSC¤,PWD6D,5,$61.       @CARD 10                    102627.00 60 000122.04 F9     102370.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.04 FA     102371.00
        CW%CDSC¤,PWD6E,5,$61.       @CARD 11                    102634.00 60 000122.04 FB     102372.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.04 FC     102373.00
        CW%CDSC¤,PWD6F,5,$61.       @CARD 12                    102641.00 60 000122.04 FD     102374.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.04 FE     102375.00
        CW%CDSC¤,PWD6G,5,$61.       @CARD 13                    102646.00 60 000122.04 FF     102376.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 00     102377.00
        CW%CDSC¤,PWD6H,5,$61.       @CARD 14                    102653.00 60 000122.05 01     102400.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 02     102401.00
        CW%CDSC¤,PWD6J,5,$61.       @CARD 15                    102660.00 60 000122.05 03     102402.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 04     102403.00
        CW%CDSC¤,PWD6K,5,$61.       @CARD 16                    102665.00 60 000122.05 05     102404.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 06     102405.00
        CW%CD¤,PWD6L,5,$61.         @CARD 17                    102672.00 20 000122.05 07     102406.00
*
*
        CW%CD¤,PRES14,130,0         @CONTROL WORD TO READ CARDS 103233.00 20 004040.00 00     102407.00
*
*       USE THE FOLLOWING CONTROL WORDS FOR PRINTOUT
*
        CW%CDSC¤,PRES1,7,$61.       @IDENTIFICATION             102677.00 60 000162.05 09     102410.00
        CW%CCR¤,PWD6,8,$61.         @WRITE AREA                 102603.00 40 000202.05 0A     102411.00
        CW%CDSC¤,PWD6B,5,$61.                                   102615.00 60 000122.05 0B     102412.00
        CW%CCR¤,PWD6,8,$61.         @WRITE AREA                 102603.00 40 000202.05 0C     102413.00
        CW%CDSC¤,PWD6C,5,$61.                                   102622.00 60 000122.05 0D     102414.00
        CW%CCR¤,PWD6,8,$61.         @WRITE AREA                 102603.00 40 000202.05 0E     102415.00
        CW%CDSC¤,PWD6D,5,$61.                                   102627.00 60 000122.05 0F     102416.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 10     102417.00
        CW%CDSC¤,PWD6E,5,$61.                                   102634.00 60 000122.05 11     102420.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 12     102421.00
        CW%CDSC¤,PWD6F,5,$61.                                   102641.00 60 000122.05 13     102422.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 14     102423.00
        CW%CDSC¤,PWD6G,5,$61.                                   102646.00 60 000122.05 15     102424.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 16     102425.00
        CW%CDSC¤,PWD6H,5,$61.                                   102653.00 60 000122.05 17     102426.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 18     102427.00
        CW%CDSC¤,PWD6J,5,$61.                                   102660.00 60 000122.05 19     102430.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 1A     102431.00
        CW%CDSC¤,PWD6K,5,$61.                                   102665.00 60 000122.05 1B     102432.00
        CW%CCR¤,PWD6,8,$61.                                     102603.00 40 000202.05 1C     102433.00
        CW%CDSC¤,PWD6L,5,$61.                                   102672.00 60 000122.05 1D     102434.00
        CW%CDSC¤,PRES2,7,$61.       @IDENTIFICATION WORD        102706.00 60 000162.05 1E     102435.00
        CW%CDSC¤,PRES14,13,$61.     @READ AREA                  103233.00 60 000322.05 1F     102436.00
        CW%CDSC¤,PRES15,13,$61.                                 103250.00 60 000322.05 20     102437.00
        CW%CDSC¤,PRES16,13,$61.                                 103265.00 60 000322.05 21     102440.00
        CW%CDSC¤,PRES17,13,$61.                                 103302.00 60 000322.05 22     102441.00
        CW%CDSC¤,PRES18,13,$61.                                 103317.00 60 000322.05 23     102442.00
        CW%CDSC¤,PRES19,13,$61.                                 103334.00 60 000322.05 24     102443.00
        CW%CDSC¤,PRES20,13,$61.                                 103351.00 60 000322.05 25     102444.00
        CW%CDSC¤,PRES21,13,$61.                                 103366.00 60 000322.05 26     102445.00
```

```
            CW%CDSC¤,PRES22,13,$61.                                  103403.00 60 000322.05 27     102446.00
            CW%CDSC¤,PRES23,13,$61.                                  103420.00 60 000322.05 28     102447.00
      *
      *
      *
      *     PUNCH TEST DATA
      *
      *     NON-ECC MODE DATA
      *
PWD1  %8¤DD%BU,8,8¤,200,004,000,040,001,000,010,000                                  200   102450.00
                                                                                      004   102450.10
                                                                                      000   102450.20
                                                                                      040   102450.30
                                                                                      001   102450.40
                                                                                      000   102450.50
                                                                                      010   102450.60
                                                                                      000   102450.70
            %8¤DD%BU,8,8¤,100,002,000,020,000,200,004,000                             100   102451.00
                                                                                      002   102451.10
                                                                                      000   102451.20
                                                                                      020   102451.30
                                                                                      000   102451.40
                                                                                      200   102451.50
                                                                                      004   102451.60
                                                                                      000   102451.70
            %8¤DD%BU,8,8¤,040,001,000,010,000,100,002,000                             040   102452.00
                                                                                      001   102452.10
                                                                                      000   102452.20
                                                                                      010   102452.30
                                                                                      000   102452.40
                                                                                      100   102452.50
                                                                                      002   102452.60
                                                                                      000   102452.70
            %8¤DD%BU,8,8¤,020,000,200,004,000,040,001,000                             020   102453.00
                                                                                      000   102453.10
                                                                                      200   102453.20
                                                                                      004   102453.30
                                                                                      000   102453.40
                                                                                      040   102453.50
                                                                                      001   102453.60
                                                                                      000   102453.70
            %8¤DD%BU,8,8¤,010,000,100,002,000,020,000,200                             010   102454.00
                                                                                      000   102454.10
                                                                                      100   102454.20
                                                                                      002   102454.30
                                                                                      000   102454.40
                                                                                      020   102454.50
                                                                                      000   102454.60
                                                                                      200   102454.70
            %8¤DD%BU,8,8¤,004,000,040,001,000,010,000,100                             004   102455.00
                                                                                      000   102455.10
                                                                                      040   102455.20
                                                                                      001   102455.30
                                                                                      000   102455.40
                                                                                      010   102455.50
                                                                                      000   102455.60
                                                                                      100   102455.70
            %8¤DD%BU,8,8¤,002,000,020,000,200,004,000,040                             002   102456.00
                                                                                      000   102456.10
                                                                                      020   102456.20
                                                                                      000   102456.30
                                                                                      200   102456.40
                                                                                      004   102456.50
                                                                                      000   102456.60
                                                                                      040   102456.70
```

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                %8¤DD%BU,8,8¤,001,000,010,000,100,002,000,020                   001  102457.00
                                                                                000  102457.10
                                                                                010  102457.20
                                                                                000  102457.30
                                                                                100  102457.40
                                                                                002  102457.50
                                                                                000  102457.60
                                                                                020  102457.70
                %8¤DD%BU,8,8¤,000,200,004,000,040,001,000,010                   000  102460.00
                                                                                200  102460.10
                                                                                004  102460.20
                                                                                000  102460.30
                                                                                040  102460.40
                                                                                001  102460.50
                                                                                000  102460.60
                                                                                010  102460.70
                %8¤DD%BU,8,8¤,000,100,002,000,020,000,200,004                   000  102461.00
                                                                                100  102461.10
                                                                                002  102461.20
                                                                                000  102461.30
                                                                                020  102461.40
                                                                                000  102461.50
                                                                                200  102461.60
                                                                                004  102461.70
                %8¤DD%BU,8,8¤,000,040,001,000,010,000,100,002                   000  102462.00
                                                                                040  102462.10
                                                                                001  102462.20
                                                                                000  102462.30
                                                                                010  102462.40
                                                                                000  102462.50
                                                                                100  102462.60
                                                                                002  102462.70
                %8¤DD%BU,8,8¤,000,020,000,200,004,000,040,001                   000  102463.00
                                                                                020  102463.10
                                                                                000  102463.20
                                                                                200  102463.30
                                                                                004  102463.40
                                                                                000  102463.50
                                                                                040  102463.60
                                                                                001  102463.70
            %8¤DD%BU,8,8¤,000,010,000,100,002,000,020,000                       000  102464.00
                                                                                010  102464.10
                                                                                000  102464.20
                                                                                100  102464.30
                                                                                002  102464.40
                                                                                000  102464.50
                                                                                020  102464.60
                                                                                000  102464.70
18      @       THE FOLLOWING ARE DATA WORDS FOR THE ECC
        @       MODE-THE CHECK BITS ARE IN OCTAL NOTATION
        @
15      @       FLOATING ZERO PATTERN                           @  C-BITS
14      PWD2    %8¤DD%BU,8,8¤,301,200,000,000,101,200,000,000   @377        301  102465.00
                                                                            200  102465.10
                                                                            000  102465.20
                                                                            000  102465.30
                                                                            101  102465.40
                                                                            200  102465.50
                                                                            000  102465.60
                                                                            000  102465.70
            %8¤DD%BU,8,8¤,350,200,200,000,230,200,200,000       @177        350  102466.00
                                                                            200  102466.10
                                                                            200  102466.20
                                                                            000  102466.30
                                                                            230  102466.40
                                                                            200  102466.50
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@       EXTENDED PUNCH TEST DATA
@
PWD4    %8¤DD%BU,8,8¤,000                   @CCB#CARRIAGE CONTROL BYTE                                  000   102513.00
        %  AZ¤DD%BU,8,8¤,PUNCH TEST USING IQS - DATA-THIS IS Z                                                 102513.10
        %  AZ¤DD%BU,8,8¤,CARD ONE OF PWD4 DATA WORDS. Z                                                      102517.50
        %  AZ¤DD%BU,8,8¤,IDENTIFIED BY A 1 IN COLUMN 80.Z                                                    102523.20
        %  AZ¤DD%BU,8,8¤,ROW 9. ECC MODEZ                                                                    102527.10
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,001                                                 000   102531.00
                                                                                                      000   102531.10
                                                                                                      000   102531.20
                                                                                                      000   102531.30
                                                                                                      000   102531.40
                                                                                                      000   102531.50
                                                                                                      000   102531.60
                                                                                                      001   102531.70
@
        %8¤DD%BU,8,8¤,000                   @CCB                                                      000   102532.00
        %  AZ¤DD%BU,8,8¤,  THIS IS CARD TWO OF PWD4 DATA Z                                                   102532.10
        %  AZ¤DD%BU,8,8¤,WORDS. IT IS IDENTIFIED WITH A 1Z                                                   102536.10
        %  AZ¤DD%BU,8,8¤, IN COLUMN 80, ROW 8...ECC MODEZ                                                    102542.10
        DD%BU,64,8¤,0                                                  000000000000000000000000   102546.00
        DD%BU,64,8¤,0                                                  000000000000000000000000   102547.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,002                                                 000   102550.00
                                                                                                      000   102550.10
                                                                                                      000   102550.20
                                                                                                      000   102550.30
                                                                                                      000   102550.40
                                                                                                      000   102550.50
                                                                                                      000   102550.60
                                                                                                      002   102550.70
@
PWD5    %8¤DD%BU,8,8¤,000                   @CCB                                                      000   102551.00
@
PWD5    %8¤DD%BU,8,8¤,000                   @CCB                                                      000   102551.00
        %  AZ¤DD%BU,8,8¤, THIS IS CARD ONE OF PWD5 DATA Z                                                    102551.10
        %  AZ¤DD%BU,8,8¤,WORDS. IT IS IDENTIFIED WITH A 1Z                                                   102555.00
        %  AZ¤DD%BU,8,8¤, IN COLUMN 78, ROW 869. NO-ECC. Z                                                   102561.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,003                                                 000   102565.00
                                                                                                      000   102565.10
                                                                                                      000   102565.20
                                                                                                      000   102565.30
                                                                                                      000   102565.40
                                                                                                      000   102565.50
                                                                                                      000   102565.60
                                                                                                      003   102565.70
        %8¤DD%BU,8,8¤,000                                                                             000   102566.00
        %  AZ¤DD%BU,8,8¤, THIS IS CARD TWO OF PWD5 DATA Z                                                    102566.10
        %  AZ¤DD%BU,8,8¤,WORDS. IT IS IDENTIFIED WITH A 1Z                                                   102572.00
        %  AZ¤DD%BU,8,8¤, IN COLUMN 78, ROW 7.NO-ECC MODEZ                                                   102576.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,004                                                 000   102602.00
                                                                                                      000   102602.10
                                                                                                      000   102602.20
                                                                                                      000   102602.30
                                                                                                      000   102602.40
                                                                                                      000   102602.50
                                                                                                      000   102602.60
                                                                                                      004   102602.70
PWD6    %8¤DD%BU,8,8¤,000                                                                             000   102603.00
        %  AZ¤DD%BU,8,8¤,XTENDED CF-1 PUNCH TESTZ                                                            102603.10
@
PWD6A   %  AZ¤DD%BU,8,8¤, CARD IS NUMBERED OCTAL IN LAST Z                                                  102606.00
        %  AZ¤DD%BU,8,8¤,COLUMN. NON-ECC MODE....Z                                                           102612.00
PWD6B   %  AZ¤DD%BU,8,8¤,CARD ONE OF EXTENDED CF-1 TEST..Z                                                  102615.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,006                                                 000   102621.00
                                                                                                      000   102621.10
                                                                                                      000   102621.20
                                                                                                      000   102621.30
                                                                                                      000   102621.40
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PWD6C   %  AZ¤DD%BU,8,8¤,CARD TWO OF EXTENDED CF-1 TEST..Z
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,007
PWD6D   %  AZ¤DD%BU,8,8¤,CARD THREE OF EXTENDED CF1 TEST.Z
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,010
PWD6E   %  AZ¤DD%BU,8,8¤,CARD FOUR OF EXTENDED CF-1 TEST.Z
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,011
PWD6F   %  AZ¤DD%BU,8,8¤,CARD FIVE OF EXTENDED CF-1 TEST.Z
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,012
PWD6G   %  AZ¤DD%BU,8,8¤,CARD SIX OF EXTENDED CF-1 TEST..Z
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,013
PWD6H   %  AZ¤DD%BU,8,8¤,CARD SEVEN OF EXTENDED CF1 TEST.Z
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,014
PWD6J   %  AZ¤DD%BU,8,8¤,CARD EIGHT OF EXTENDED CF1 TEST.Z
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,015
```

```
000 102621.50
000 102621.60
006 102621.70
    102622.00
000 102626.00
000 102626.10
000 102626.20
000 102626.30
000 102626.40
000 102626.50
000 102626.60
007 102626.70
    102627.00
000 102633.00
000 102633.10
000 102633.20
000 102633.30
000 102633.40
000 102633.50
000 102633.60
010 102633.70
    102634.00
000 102640.00
000 102640.10
000 102640.20
000 102640.30
000 102640.40
000 102640.50
000 102640.60
011 102640.70
    102641.00
000 102645.00
000 102645.10
000 102645.20
000 102645.30
000 102645.40
000 102645.50
000 102645.60
012 102645.70
    102646.00
000 102652.00
000 102652.10
000 102652.20
000 102652.30
000 102652.40
000 102652.50
000 102652.60
013 102652.70
    102653.00
000 102657.00
000 102657.10
000 102657.20
000 102657.30
000 102657.40
000 102657.50
000 102657.60
014 102657.70
    102660.00
000 102664.00
000 102664.10
000 102664.20
000 102664.30
000 102664.40
000 102664.50
000 102664.60
015 102664.70
```

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PWD6K   %  AZ¤DD%BU,8,8¤,CARD NINE OF EXTENDED CF-1 TEST.Z                                  102665.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,016                          000  102671.00
                                                                               000  102671.10
                                                                               000  102671.20
                                                                               000  102671.30
                                                                               000  102671.40
                                                                               000  102671.50
                                                                               000  102671.60
                                                                               016  102671.70
PWD6L   %  AZ¤DD%BU,8,8¤,CARD TEN OF EXTENDED CF-1 TEST..Z                                 102672.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,017                          000  102676.00
                                                                               000  102676.10
                                                                               000  102676.20
                                                                               000  102676.30
                                                                               000  102676.40
                                                                               000  102676.50
                                                                               000  102676.60
                                                                               017  102676.70
PRES1   %8¤DD%BU,8,8¤,001                                                      001  102677.00
        %  AZ¤DD%BU,8,8¤,THIS IS THE DATA FROM THE WRITEZ                           102677.10
        %  AZ¤DD%BU,8,8¤, AREA OF THE PUNCH TEST.Z                                  102703.00
PRES2   %8¤DD%BU,8,8¤,001                                                      001  102706.00
        %  AZ¤DD%BU,8,8¤,THIS IS THE DATA FROM THE READ Z                           102706.10
        %  AZ¤DD%BU,8,8¤, AREA OF THE PUNCH TEST.Z                                  102712.00
PRES3   DR%BU,64,8¤,30                  @READ-IN AREA          36.00             102715.00
PRES3A  DR%BU,64,8¤,26                  @READ-IN AREA-ECC      32.00             102753.00
PRES4   DR%BU,64,8¤,15                  @NON-ECC MODE          17.00             103005.00
PRES5   DR%BU,64,8¤,15                                         17.00             103024.00
PRES6   DR%BU,64,8¤,15                                         17.00             103043.00
PRES7   DR%BU,64,8¤,15                                         17.00             103062.00
PRES8   DR%BU,64,8¤,15                                         17.00             103101.00
PRES9   DR%BU,64,8¤,15                                         17.00             103120.00
PRES10  DR%BU,64,8¤,15                                         17.00             103137.00
PRES11  DR%BU,64,8¤,15                                         17.00             103156.00
PRES12  DR%BU,64,8¤,15                                         17.00             103175.00
PRES13  DR%BU,64,8¤,15                                         17.00             103214.00
PRES14  DR%BU,64,8¤,13                  @READ-IN AREA          15.00             103233.00
PRES15  DR%BU,64,8¤,13                  @NON-ECC MODE          15.00             103250.00
PRES16  DR%BU,64,8¤,13                                         15.00             103265.00
PRES17  DR%BU,64,8¤,13                                         15.00             103302.00
PRES18  DR%BU,64,8¤,13                                         15.00             103317.00
PRES19  DR%BU,64,8¤,13                                         15.00             103334.00
PRES20  DR%BU,64,8¤,13                                         15.00             103351.00
PRES21  DR%BU,64,8¤,13                                         15.00             103366.00
PRES22  DR%BU,64,8¤,13                                         15.00             103403.00
PRES23  DR%BU,64,8¤,13                                         15.00             103420.00
END     DR%BU,64,8¤,1                                           1.00             103435.00
        END,START                                          100000.00             103436.00
```

SLC,64,0			000100.00
PUNID,BX0-RDR		BX0-RDR	
END,64,0		100.00	000100.00

PRNID,BX0 RDR READER TEST DECK 1 ....

```
          SLC,64,0                                                                    000100.00
@         SLC VALUE IS MEANINGLESS
          PUNFUL
@         THIS IS TEST PATTERN DECK 1. RESULTS CAN BE
@         EASILY DETERMINED BY USING CHKRDR CONTROL WORD
@         SEQUENCE,AND/OR DATA DISPLAY.
@         A SECOND PATTERN DECK WILL BE MADE AVAILABLE
@         WHICH CONTAINS SPECIAL PATTERN CARDS.CHECKS ARE
@         MADE BY DISPLAYING MEMORY.USE TEST DECK 2 IF
@         PRINTER OPERATION IS DOUBTFUL.....
@
@
CARD1     %8¤DD%BU,8,8¤,000                                                 000   000100.00
          %  AZ¤DD%BU,8,8¤,CARD1 .....FIRST CARD READ...DATA IS INZ                 000100.10
          %  AZ¤DD%BU,8,8¤, IQS FORMAT. WORD COUNT ON READ WAS 15 .Z                 000105.00
          %  AZ¤DD%BU,8,8¤,READER PATTERNS IN LATER TEST...Z                         000112.00
          %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,001  @CRD 1 IDENT        000   000116.00
                                                                             000   000116.10
                                                                             000   000116.20
                                                                             000   000116.30
                                                                             000   000116.40
                                                                             000   000116.50
                                                                             000   000116.60
                                                                             001   000116.70
@
CARD2     %8¤DD%BU,8,8¤,000                                                 000   000117.00
          %  AZ¤DD%BU,8,8¤,WDCT1  Z                                                 000117.10
          %  AZ¤DD%BU,8,8¤,FAILURE Z                                                000120.00
          %  AZ¤DD%BU,8,8¤,IF THIS PRINTS OR IS IN MEM, WD Z                        000121.00
          %  AZ¤DD%BU,8,8¤,CNT-1-WAS NOT HANDLED BY BX.....Z                        000125.00
          DR%BU,64,8¤,4              @WORDS 11-14 CARD 2 ARE ZERO.        4.00     000131.00
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```
                %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,002  @CRD 2 IDENT                    000   000135.00
                                                                                               000   000135.10
                                                                                               000   000135.20
                                                                                               000   000135.30
                                                                                               000   000135.40
                                                                                               000   000135.50
                                                                                               000   000135.60
                                                                                               002   000135.70
          @
CARD3     %8¤DD%BU,8,8¤,000          @                                                     000   000136.00
          %  AZ¤DD%BU,8,8¤,WORD COUNT -2- Z                                                       000136.10
          %  AZ¤DD%BU,8,8¤,FAILURE Z                                                            000140.00
          DR%BU,64,8¤,11             @CARD 3 BLANK LOCATIONS          13.00                  000141.00
          %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,003  @CRD 3 IDENT                    000   000154.00
                                                                                               000   000154.10
                                                                                               000   000154.20
                                                                                               000   000154.30
                                                                                               000   000154.40
                                                                                               000   000154.50
                                                                                               000   000154.60
                                                                                               003   000154.70
          @
CARD4     %8¤DD%BU,8,8¤,000                                                                 000   000155.00
          %  AZ¤DD%BU,8,8¤,IF THIS PRINTS SKIP FLAG FAILEDZ                                      000155.10
          %  AZ¤DD%BU,8,8¤,CARD4-SKIP FLAG TEST....Z                                             000161.00
          DR%BU,64,8¤,7                                        7.00                              000164.00
          %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,004  @CRD 4 IDENT                    000   000173.00
                                                                                               000   000173.10
                                                                                               000   000173.20
                                                                                               000   000173.30
                                                                                               000   000173.40
                                                                                               000   000173.50
                                                                                               000   000173.60
```

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                                                                                    004   000173.70

@
CARD5   %8¤DD%BU,8,8¤,000                                                           000   000174.00
        %  AZ¤DD%BU,8,8¤,CARD 5 THREE CARD READ, ABCDEFGHIJKLMNOZ                          000174.10
        %  AB¤DD%BU,8,8¤,PQRSTUVWXYZ 0123456789 CARD 5 THREE CARDB                         000201.00
        %  A9¤DD%BU,8,8¤, READ ABCDEFGHIJKLMNOPQRSTUVWXYZ9                                 000206.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,005  @CRD 5 IDENT                000   000212.00
                                                                                    000   000212.10
                                                                                    000   000212.20
                                                                                    000   000212.30
                                                                                    000   000212.40
                                                                                    000   000212.50
                                                                                    000   000212.60
                                                                                    005   000212.70

@
CARD6   %8¤DD%BU,8,8¤,000                                                           000   000213.00
        %  AZ¤DD%BU,8,8¤,CARD 6 ...3 CARD READ, CARD 2  HIJKLMNOZ                          000213.10
        %  AB¤DD%BU,8,8¤,PQRSTUVWXYZ 0123456789 CARD 6 THREE CARDB                         000220.00
        %  A9¤DD%BU,8,8¤, READ ABCDEFGHIJKLMNOPQRSTUVWXYZ9                                 000225.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,006  @CRD 6 IDENT                000   000231.00
                                                                                    000   000231.10
                                                                                    000   000231.20
                                                                                    000   000231.30
                                                                                    000   000231.40
                                                                                    000   000231.50
                                                                                    000   000231.60
                                                                                    006   000231.70

@
CARD7   %8¤DD%BU,8,8¤,000                                                           000   000232.00
        %  AZ¤DD%BU,8,8¤,CARD 7 ...3 CARD READ, CARD 3  HIJKLMNOZ                          000232.10
        %  AB¤DD%BU,8,8¤,PQRSTUVWXYZ 0123456789 CARD 7 THREE CARDB                         000237.00
        %  A9¤DD%BU,8,8¤, READ ABCDEFGHIJKLMNOPQRSTUVWXYZ9                                 000244.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,007  @CRD 7 IDENT                000   000250.00
```

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                                                                                                                    000   000250.10
                                                                                                                    000   000250.20
                                                                                                                    000   000250.30
                                                                                                                    000   000250.40
                                                                                                                    000   000250.50
                                                                                                                    000   000250.60
                                                                                                                    007   000250.70
*
CARD8   %8¤DD%BU,8,8¤,000                                                                                           000   000251.00
        %  AZ¤DD%BU,8,8¤,CARD 8 TEN CARD READ.8388888888888888Z                                                          000251.10
        %  AZ¤DD%BU,8,8¤,88888888888888888888888888888888888888888888Z                                                 000256.00
        %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 1 Z                                                                   000263.00
        DR%BU,64,8¤,1                                                                  1.00                           000266.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,010  @CRD 8 IDENT                                                  000   000267.00
                                                                                                                    000   000267.10
                                                                                                                    000   000267.20
                                                                                                                    000   000267.30
                                                                                                                    000   000267.40
                                                                                                                    000   000267.50
                                                                                                                    000   000267.60
                                                                                                                    010   000267.70
*
CARD9   %8¤DD%BU,8,8¤,000                                                                                           000   000270.00
        %  AZ¤DD%BU,8,8¤,CARD 9 TEN CARD READ. 999999999999999Z                                                          000270.10
        %  AZ¤DD%BU,8,8¤,999999999999999999999999999999999999999999Z                                                   000275.00
        %  AZ¤DD%BU,8,8¤, -TEN CARD READ.-CARD 2 Z                                                                   000302.00
        DR%BU,64,8¤,1                                                                  1.00                           000305.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,011  @CRD 9 IDENT                                                  000   000306.00
                                                                                                                    000   000306.10
                                                                                                                    000   000306.20
                                                                                                                    000   000306.30
                                                                                                                    000   000306.40
                                                                                                                    000   000306.50
```

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                                                                                                                    000   000306.60
                                                                                                                    011   000306.70
*
CARD10  %8¤DD%BU,8,8¤,000                                                                                           000   000307.00 6
        %  AZ¤DD%BU,8,8¤,CARD 10 TEN CARD READ. Z                                                                        000307.10
        DR%BU,64,8¤,7                                                                    7,00                              000312.00
        %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 3 Z                                                                    000321.00
        DR%BU,64,8¤,1                                                                    1,00                              000324.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,012  @CRD 10 IDENT                                                000   000325.00
                                                                                                                    000   000325.10
                                                                                                                    000   000325.20
                                                                                                                    000   000325.30
                                                                                                                    000   000325.40
                                                                                                                    000   000325.50
                                                                                                                    000   000325.60
                                                                                                                    012   000325.70
*
CARD11  %8¤DD%BU,8,8¤,000                                                                                           000   000326.00 6
        %  AZ¤DD%BU,8,8¤,CARD 11 TEN CARD READ. Z                                                                        000326.10
        DR%BU,64,8¤,7                                                                    7,00                              000331.00
        %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 4 Z                                                                    000340.00
        DR%BU,64,8¤,1                                                                    1,00                              000343.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,013  @CRD 11 IDENT                                                000   000344.00
                                                                                                                    000   000344.10
                                                                                                                    000   000344.20
                                                                                                                    000   000344.30
                                                                                                                    000   000344.40
                                                                                                                    000   000344.50
                                                                                                                    000   000344.60
                                                                                                                    013   000344.70
*
CARD12  %8¤DD%BU,8,8¤,000                                                                                           000   000345.00 6
        %  AZ¤DD%BU,8,8¤,CARD 12 TEN CARD READ. Z                                                                        000345.10
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                DR%BU,64,8¤,7                                                   7.00                    000350.00
                %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 5 Z                                             000357.00
                DR%BU,64,8¤,1                                                   1.00                    000362.00
                %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,014  @CRD 12 IDENT                    000     000363.00
                                                                                                000     000363.10
                                                                                                000     000363.20
                                                                                                000     000363.30
                                                                                                000     000363.40
                                                                                                000     000363.50
                                                                                                000     000363.60
                                                                                                014     000363.70
                @
CARD13          %8¤DD%BU,8,8¤,000                                                               000     000364.00 6
                %  AZ¤DD%BU,8,8¤,CARD 13 TEN CARD READ. Z                                               000364.10
                DR%BU,64,8¤,7                                                   7.00                    000367.00
                %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 6 Z                                              000376.00
                DR%BU,64,8¤,1                                                   1.00                    000401.00
                %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,015  @CRD 13 IDENT                    000     000402.00
                                                                                                000     000402.10
                                                                                                000     000402.20
                                                                                                000     000402.30
                                                                                                000     000402.40
                                                                                                000     000402.50
                                                                                                000     000402.60
                                                                                                015     000402.70
                @
CARD14          %8¤DD%BU,8,8¤,000                                                               000     000403.00 6
                %  AZ¤DD%BU,8,8¤,CARD 14 TEN CARD READ. Z                                               000403.10
                DR%BU,64,8¤,7                                                   7.00                    000406.00
                %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 7 Z                                              000415.00
                DR%BU,64,8¤,1                                                   1.00                    000420.00
                %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,016  @CRD 14 IDENT                    000     000421.00
                                                                                                000     000421.10
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                                                                                                                    000   000421.20
                                                                                                                    000   000421.30
                                                                                                                    000   000421.40
                                                                                                                    000   000421.50
                                                                                                                    000   000421.60
                                                                                                                    016   000421.70
*
CARD15  %8¤DD%BU,8,8¤,000                                                                                           000   000422.00 6
        %  AZ¤DD%BU,8,8¤,CARD 15 TEN CARD READ. Z                                                                        000422.10
        DR%BU,64,8¤,7                                                                    7.00                               000425.00
        %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 8 Z                                                                       000434.00
        DR%BU,64,8¤,1                                                                    1.00                               000437.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,017  @CRD 15 IDENT                                                   000   000440.00
                                                                                                                    000   000440.10
                                                                                                                    000   000440.20
                                                                                                                    000   000440.30
                                                                                                                    000   000440.40
                                                                                                                    000   000440.50
                                                                                                                    000   000440.60
                                                                                                                    017   000440.70
*
CARD16  %8¤DD%BU,8,8¤,000                                                                                           000   000441.00 6
        %  AZ¤DD%BU,8,8¤,CARD 16 TEN CARD READ. Z                                                                        000441.10
        DR%BU,64,8¤,7                                                                    7.00                               000444.00
        %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 9 Z                                                                       000453.00
        DR%BU,64,8¤,1                                                                    1.00                               000456.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,021  @CRD 16 IDENT                                                   000   000457.00
                                                                                                                    000   000457.10
                                                                                                                    000   000457.20
                                                                                                                    000   000457.30
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                                                                                                                    000   000457.50
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                                                                                                                    021   000457.70

*
CARD17  %8¤DD%BU,8,8¤,000                                                                                           000   000460.00 6
        %  AZ¤DD%BU,8,8¤,CARD 17 TEN CARD READ. Z                                                                        000460.10
        DR%BU,64,8¤,7                                                                  7.00                                000463.00
        %  AZ¤DD%BU,8,8¤,  TEN CARD READ.-CARD 10Z                                                                       000472.00
        DR%BU,64,8¤,1                                                                  1.00                                000475.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,022  @CRD 17 IDENT                                                   000   000476.00
                                                                                                                    000   000476.10
                                                                                                                    000   000476.20
                                                                                                                    000   000476.30
                                                                                                                    000   000476.40
                                                                                                                    000   000476.50
                                                                                                                    000   000476.60
                                                                                                                    022   000476.70

*
CARD18  %8¤DD%BU,8,8¤,000                                                                                           000   000477.00 6
        %  AZ¤DD%BU,8,8¤,CARD 18. TWO CARD READ WITH MF-0. ONLY Z                                                        000477.10
        %  AZ¤DD%BU,8,8¤,ONE CARD SHOULD READ....Z                                                                       000504.00
        DR%BU,64,8¤,6                                                                  6.00                                000507.00
        %8¤DD%BU,8,8¤,000,000,000,000,000,000,000,023  @CRD 18 IDENT                                                   000   000515.00
                                                                                                                    000   000515.10
                                                                                                                    000   000515.20
                                                                                                                    000   000515.30
                                                                                                                    000   000515.40
                                                                                                                    000   000515.50
                                                                                                                    000   000515.60
                                                                                                                    023   000515.70

*
CARD19  %8¤DD%BU,8,8¤,000                                                                                           000   000516.00 6
        %  AZ¤DD%BU,8,8¤,THIS CARD SHOULD NOT BE READ   Z                                                               000516.10
        %  AZ¤DD%BU,8,8¤,CARD 19 Z                                                                                       000522.00
        DR%BU,64,8¤,10                                                                 12.00                               000523.00
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%8¤DD%8U,8,8¤,000,000,000,000,000,000,000,024 @CRD 19 IDENT		000	000535.00
		000	000535.10
		000	000535.20
		000	000535.30
		000	000535.40
		000	000535.50
		000	000535.60
		024	000535.70
NOP	0.30.00		000536.00
END,64.0	100.00		000536.40